Beyond the Jack-of-All-Trades Hypothesis: The Importance of Business and Specialized Skills for Business Ownership Entry and Success

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October 30, 2026

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Abstract

Lazear's well-known jack-of-all-trades (JOAT) model (2004, 2005) hypothesizes that business owners require balanced skills—an argument supported by numerous empirical studies, typically by showing that the number of prior occupations or employers held by workers is positively associated with business ownership. I extend the JOAT literature by arguing that (a) business owners do not need a balance of *all* skills, but instead need a set of *business* skills plus specialized skills relevant to their business; (b) business skills are likely to be more important for incorporated, capital-intensive, "advanced" businesses than for "basic" businesses; and (c) evidence in support of the JOAT hypothesis is likely to be sensitive to whether skill balance is measured directly or via orthodox, prior mobility measures. Using NLSY79 work histories linked to O*NET data to measure business skills, specialized skills, and skill balance, I find that all three arguments are supported by the data: business skills are strong predictors of entry into advanced (but not basic) business ownership; both business and specialized skills are associated with higher revenue and earnings for all business owners; and my direct measure of skill balance is unassociated with all three outcomes, whereas an orthodox (occupation count) measure strongly predicts entry into basic business ownership largely because it reflects involuntary job loss. In short, I find that business owners benefit from being jacks of business trades rather than jacks of all trades.

JEL codes: L26, J24, J62, M13

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1 Introduction

What kind of work experience prepares someone to run a business? The best-known answer comes from Lazear's (2004, 2005) jack-of-all-trades (JOAT) model, which posits that because entrepreneurs must handle many functions at once, workers with balanced skills have a comparative advantage in entrepreneurship, while workers with specialized skills have a comparative advantage in wage employment.[1] An extensive empirical literature finds evidence in support of this hypothesis, typically by using regression analysis to relate a measure of skill balance to the probability of becoming an entrepreneur (Wagner, 2003, 2006; Åstebro and Thompson, 2011; Aldén, Hammarstedt and Neuman, 2017). Skill balance is usually proxied by the number of occupations or employers a worker has held, and entrepreneurship is defined as any self-employment and/or business ownership.[2]

I extend the JOAT literature by introducing three innovations: First, I argue that what prepares a worker for business ownership is not balance across *all* skills, but rather two types of skills relevant to establishing and operating a business: *business skills*, meaning core business knowledge, managerial skills, and interpersonal skills, and *specialized skills*, meaning expertise directly relevant to the business. This distinction is already present in an example provided by Lazear: "Consider the founder of a new small restaurant. In addition to being a good cook, the founder must be able to obtain funds, hire workers, choose location and decor, obtain food supplies at a reasonable cost, keep books, and market the restaurant" (Lazear, 2004, p. 208). Although Lazear implicitly suggests with this example that workers need a balance of business skills and specialized (cooking) skills, in subsequent empirical work (including Lazear's (2005) own test) this distinction is lost: researchers typically measure skill balance by counting past occupations or employers, which implicitly gives each occupation or employer equal weight. For example, a worker who was a truck driver, a chef, and a manager before opening a restaurant is considered to have more skill balance than a worker who spent the same amount of time as a chef and a manager, even though truck driving is unlikely to provide skills relevant to running a restaurant. By measuring each worker's business skills and specialized skills, I focus on the types of skills most relevant to business ownership.

Second, I distinguish between two types of business ownership. Most studies pool all self-

[1] Even those entrepreneurs who put together teams to launch a business require balanced skills to select and manage the team.

[2] Although Lazear (2004, 2005) uses the term "entrepreneur" in his work, he has in mind owners of relatively complex businesses (including, for example, restaurants) who may or may not be innovators as envisioned by Schumpeter (1912) in his widely-used definition of an entrepreneur. I follow the majority of JOAT-related studies in considering anyone working for their own business to be a business owner; because these workers are reported in my data as self-employed or business owners or both (see section 4), I use the terms "self-employment" and "business ownership" interchangeably throughout this paper.

employment and/or business ownership, which means, for example, that a freelance house cleaner and the founder of an incorporated construction firm fall into the same category even though they perform different functions and may require different skills. Both the JOAT hypothesis and my argument that business skills are what matter are likely to apply to businesses that are complex enough to require coordinating multiple functions, such as the restaurant in Lazear's example. Pooling these businesses with simpler businesses can obscure the roles that different skills play. I classify businesses that are incorporated and/or start with higher levels of capital as *advanced* businesses, where owners are more likely to manage functions such as financing, hiring, and marketing in addition to using their specialized skills; all other businesses are classified as *basic* businesses, where owners are expected to rely primarily on specialized skills. My definition of advanced businesses appears to conform to Lazear's (2004, 2005) definition of an "entrepreneur" (see footnote 2).

Third, I revisit tests of the JOAT hypothesis by comparing a conventional occupation count measure with a direct measure of skill balance. Occupation count can be a poor proxy for skill balance for two reasons. It treats every new occupation as an equal step toward balance regardless of how similar the occupation is to past occupations; for example, a worker who moves from "carpenter" to "cabinetmaker" adds an occupation to the count, even though both build similar skills. Moreover, the count also rises with job churning, such as repeated involuntary job losses, which have little to do with skill accumulation. By controlling for occupation count, a direct measure of skill balance, and a count of involuntary job losses, I can assess whether occupation count predicts entry into business ownership because it captures skill balance, or because it reflects job churning.

Before pursuing these three innovations empirically, I present a simple, static choice model to clarify how business skills and specialized skills affect workers' choices among wage employment, basic business ownership, and advanced business ownership, as well as their success in each type of business. I then turn to an empirical analysis in which I combine individuals' decades-long work histories from the 1979 National Longitudinal Survey of Youth (NLSY79) with occupation-specific skill data from the Occupational Information Network (O*NET). I use these data to estimate a competing-risks model of workers' year-by-year transitions from wage employment or nonemployment into either basic or advanced business ownership. I also examine subsequent success in each type of business, measured by business revenue and earnings. The key regressors are measures of business skills and specialized skills, which I construct by weighting O*NET's skill and knowledge ratings of each occupation a worker has held by the years spent in that occupation. I also construct a direct measure of skill balance based on how evenly a worker's accumulated skills are spread across different types of skills and knowledge; I use this measure along with the conventional occupation

count in the business ownership entry model to replicate and extend tests of the JOAT hypothesis.[3]

Estimation of these empirical models demonstrates that my three extensions of the JOAT literature are warranted. I find that (a) balance across all skills does not matter for business ownership, but both business skills and specialized skills *do* matter; (b) the estimated effects of all skill measures differ between basic and advanced businesses; and (c) a count of past occupations held—which is often used as a proxy for skill balance in tests of the JOAT hypothesis—reflects job churning more than skill balance.

Specifically, I find that a half-standard-deviation increase in business skills is predicted to increase the probability of entering advanced business ownership by 34%, and to increase business revenue of advanced business owners by 42%. In contrast, the same increment in business skills is predicted to lower the probability of entering basic business ownership by 14%, and to increase basic business owners' revenue by only 25%. Specialized skills are negatively associated with entry into both types of business (because they are even more valuable in wage employment), but a half-standard-deviation increase in specialized skills is predicted to increase advanced business revenue by an imprecisely-estimated 7% and increase basic revenue by 16%. In short, all business owners benefit from specialized skills and from being jacks of business trades, but the payoffs to these skills differ by business type.

In addition, my direct, O*NET-based measure of skill balance proves to be unrelated to the probability of entering either basic and advanced business ownership once business skills and specialized skills are accounted for, and to have little association with subsequent revenue and earnings of business owners of either type. I also find that the number of past occupations positively predicts entry into basic business ownership but not advanced business ownership, and that about a third of this relationship is explained by involuntary job loss. Together, these findings demonstrate that skill balance, when measured directly, is not an important determinant of entry into business ownership or subsequent success, in contrast to the JOAT model's prediction. Moreover, my findings suggest that conventional tests of the JOAT hypothesis conflate the role of skill balance with the roles of business skills and job churning, while obscuring important differences between types of business ownership.

My findings have implications for policies that promote business ownership. They suggest that efforts to prepare prospective business owners should focus on building business

[3]I cannot completely separate a direct (causal) effect of business skills and specialized skills from the effects of other factors (e.g., wealth, risk tolerance, a preference for autonomy) that affect skill acquisition via occupational choice and also affect entry into business ownership and success outcomes. I use a "selection on observables" strategy by controlling for measures of wealth, risk tolerance, cognitive ability, personality traits, family background, and other factors. I describe this approach (as well as my inability to use instrumental variables estimation) in section 5.

skills, such as managerial, financial, and interpersonal skills, rather than a broad balance of skills. Programs supported by the U.S. Small Business Administration (SBA), such as Small Business Development Centers (SBDCs) and SCORE, already take this approach by offering business training and mentoring to prospective and existing owners.[4] Owners of basic businesses might also benefit from assistance in building specialized skills because these skills are strongly associated with business revenue and earnings.

2 Related Literature

Lazear's (2004, 2005) jack-of-all-trades (JOAT) hypothesis formalizes the idea that business owners benefit from being generalists because they need to handle many tasks, such as marketing, operations, finance, and personnel management. Using data on Stanford MBA graduates, Lazear (2005) finds that individuals who have worked in more roles and studied more fields are indeed more likely to become business owners.

Subsequent to Lazear's (2004, 2005) work, many other researchers have tested the JOAT hypothesis in the context of entry into business ownership using a variety of measures of skill balance. Most studies proxy skill balance with counts of the employers, occupations, industries, or functional areas (corporate departments such as marketing, finance, and R&D) that a worker has experienced since starting her career (Wagner, 2003, 2006; Silva, 2007; Åstebro and Thompson, 2011; Stuetzer, Obschonka and Schmitt-Rodermund, 2013; Lechmann and Schnabel, 2014; Chen and Thompson, 2016; Tegtmeier, Kurczewska and Halberstadt, 2016; Sorgner and Fritsch, 2018), while others use the variation in scores across a set of pre-labor market, skill-specific aptitude tests (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017). Using these measures, existing studies generally find that individuals with more balanced skills are more likely to enter business ownership, and Stuetzer, Goethner and Cantner (2012) find that start-up projects whose founders have more balanced skills make more progress toward founding a business. Exceptions include Hartog, Van Praag and Van Der Sluis (2010), who find that skill balance is positively associated with business earnings but not entry, and Silva (2007), who finds that the relationship between skill balance and business ownership is no longer statistically significant after controlling for individual fixed effects, suggesting that persistent individual characteristics may account for part of the observed association.

A smaller literature examines post-entry business outcomes—typically earnings or survival—using the measures of skill balance described above. Several such studies find that balanced

[4]SCORE originally stood for Service Corps of Retired Executives. It is a nonprofit organization that provides mentoring to small business owners, with mentors now including both active and retired business professionals.

skills are associated with higher earnings among business owners than among wage workers (Hartog, Van Praag and Van Der Sluis, 2010; Åstebro, Chen and Thompson, 2011; Bublitz and Noseleit, 2014; Aldén, Hammarstedt and Neuman, 2017). An exception is Åstebro and Thompson (2011), who find that business owners who have worked in more occupations and industries have lower household income than those who have worked in fewer. Studies that focus on business survival typically find that business owners with balanced skills tend to stay in business longer (Aldén, Hammarstedt and Neuman, 2017; Kurczewska and Mackiewicz, 2020). Although business revenue has received relatively little attention, one exception is Aldén, Hammarstedt and Neuman (2017), who examine not only survival and earnings but also revenue and profits, and find that all four outcomes are associated with balanced skills.

A related literature suggests that the types of skills workers accumulate affect business ownership, beyond the balance of skills. For example, Unger et al. (2011) and Ganotakis (2012) find that experience in running a business, such as start-up, management, and commercial experience, is associated with larger, faster-growing, and more profitable businesses. Koster and Andersson (2018) find that relevant prior knowledge gained through related occupations predicts higher earnings and longer business survival, while Hvide and Oyer (2025) find that early industry exposure predicts greater business survival, employment, and profitability. Within the JOAT literature, Lechmann and Schnabel (2014) find that business ownership requires more expert-level skills than does wage work. Chen and Thompson (2016) account for specific types of experience, such as finance, business administration, and senior management, and find that they largely explain the positive relationship between skill balance and the probability of founding a business. However, they measure each type of experience by whether a worker has ever held a job in that occupation.

A number of studies focus, as I do, on the fact that businesses are highly heterogeneous. Hurst and Pugsley (2011) show that most businesses are small and have no plans to grow or innovate, and argue that pooling all businesses can be inappropriate for testing standard theories of business ownership. Levine and Rubinstein (2017, 2018) use NLSY79 data to show that incorporated business owners perform more nonroutine cognitive tasks and earn more than wage workers, whereas unincorporated business owners perform more manual tasks and earn less; they also show that incorporated businesses start with substantially more capital than unincorporated businesses. Light and Munk (2018) find that much of the self-employment reported in the NLSY79 consists of contract or home-based work that respondents do not separately report as a business that they own, while Humphries (2017) finds that workers with relatively weak labor market attachment are more likely than others to run short-lived, low-capital, unincorporated businesses. This evidence suggests that the skills relevant for entry into and success in business ownership may depend on the type of

business, yet the JOAT literature rarely distinguishes between these types.[5]

Finally, the occupation and employer counts commonly used in the JOAT literature measure job mobility, which is interpreted as a proxy for skill balance under the assumption that workers who move across occupations or employers accumulate more balanced skills than do their less mobile counterparts. However, this assumption is unlikely to hold for most workers. Mobility might reflect productive job shopping (Neal, 1999), but it might also reflect poor performance and involuntary job loss (Groes, Kircher and Manovskii, 2015). This second kind of mobility, which I refer to as job churning, is itself associated with entry into business ownership. In particular, workers who change jobs frequently or experience unemployment are more likely to become business owners (Evans and Leighton, 1989; Åstebro, Chen and Thompson, 2011), and those with poor labor market prospects tend to start small, short-lived businesses (Poschke, 2013). As a result, counts of occupations or employers may predict entry into business ownership even if skill balance plays no role. Studies that use the dispersion of ability test scores are not affected by job churning, but they examine a different concept: the balance of abilities measured before labor market entry (Hartog, Van Praag and Van Der Sluis, 2010; Aldén, Hammarstedt and Neuman, 2017).

I build on these studies in three ways. First, I distinguish business skills from specialized skills and offer a decision-making framework that shows how each relates to entry into business ownership and, independently, to subsequent business revenue or earnings. Following prior studies that combine workers' job histories with data on the skill requirements of occupations (Gathmann and Schönberg, 2010; Lise and Postel-Vinay, 2020), I measure how much of each type of skill a worker has accumulated across all her jobs, rather than whether she has ever held a job of that uses that skill. Second, rather than pooling all business owners, I distinguish between two types of business ownership, basic and advanced, which I define on the basis of both incorporation status and start-up capital. Third, my use of a direct measure of skill balance based on workers' occupational histories combined with O*NET data on the skill content of each occupation is an important contribution to the JOAT literature. In addition, my strategy of combining this direct measure with an occupation count measure (as well as a measure of prior involuntary job loss) is designed to reveal whether conventional measures of skill balance provide support for the JOAT hypothesis or simply reflect churning.

[5]Lechmann and Schnabel (2014) are an exception: they distinguish between business owners with and without employees and find that the positive relationship between the number of occupations and business ownership holds only among owners without employees.

3 Conceptual Framework

In this section, I present a static choice model to formalize how a worker's business skills and specialized skills (my two key regressors) affect her choice among wage employment, basic business ownership, and advanced business ownership, and how these skills affect her payoffs conditional on entry. The model provides a conceptual framework for understanding how business skills and specialized skills might differently predict entry into each type of business ownership and subsequent business revenue and earnings. I begin with a comparison of my model to Lazear's (2004, 2005) JOAT model, which forms the foundation for my approach.

3.1 Model Setup and Comparison to Lazear's Jack-of-All-Trades Model

In Lazear (2004, 2005), a worker has two arbitrary skills, x_1 and x_2, and chooses between working as a specialist in wage employment and becoming a business owner. A specialist is paid for her stronger skill, so her payoff is $\max(x_1, x_2)$. A business owner must perform many tasks herself, so her payoff is limited by her weaker skill; that is, her payoff is $\lambda \min(x_1, x_2)$, where λ is the market value of a business owner's talent. She becomes a business owner when $\lambda \min(x_1, x_2) > \max(x_1, x_2)$, which is most likely to hold when x_1 and x_2 are at similar levels. As a result, workers with balanced skills become business owners, and workers with one strong skill remain specialists in wage employment.

My framework keeps the basic structure of Lazear's model, in which a worker compares the payoff of wage employment with the payoff of business ownership. I follow Lazear in assuming that skills are given, but the two skills are not arbitrary. They are business skills, BX, and specialized skills, SX, the two types of skills that running a business likely requires, and the worker accumulates them through prior work experience.[6] Unlike Lazear, I distinguish between two types of business ownership that pay different returns to each skill. Like Lazear, I abstract from nonemployment, so the worker chooses one of three employment options, $j \in \{W, B, A\}$: wage employment (W), basic business ownership (B), or advanced business ownership (A).[7]

I define the payoff y_j as the economic return on the worker's skills that she receives from option j. In wage employment, this is her earnings, and in business ownership, it is the income her business generates, whether she takes it home or reinvests it in the business. In

[6] Because my goal is to show how given skills affect the choice, I do not model how workers accumulate skills. Relaxing this assumption is an area for future research. For example, an occupational choice or lifetime skill investment model could be extended to include business ownership as an option, so that workers decide how much of each skill to accumulate while knowing that they might later start a business.

[7] In the data, the alternative to business ownership pools wage employment and nonemployment (section 4.2); however, 56% of entries into basic business ownership and 68% of entries into advanced business ownership occur while the worker still holds a wage job or within one month of its end date.

Lazear's model, specialists are paid only for their stronger skill and business owners only for their weaker skill, which makes the balance between skills central to the choice. In my framework, each skill contributes to the payoff in every option, but at a rate that differs across options. Let r_j^{BX} and r_j^{SX} denote the returns to BX and SX in option j. I normalize both returns in wage employment to one, so the returns in business ownership are measured relative to wage employment. The payoffs are additive in the two skills as follows:[8]

$$y_W = BX + SX, \tag{1}$$

$$y_B = r_B^{BX}\, BX + r_B^{SX}\, SX, \tag{2}$$

$$y_A = r_A^{BX}\, BX + r_A^{SX}\, SX. \tag{3}$$

Skills are not the only individual factors that determine whether a worker becomes a business owner. Workers also differ in wealth, risk tolerance, personality traits, preference for autonomy, business aspirations, and many other characteristics that affect the relative value of business ownership, independent of skill levels. I summarize all of these factors in two parameters, τ_B and τ_A (after normalizing the taste for wage employment to $\tau_W = 0$), which I refer to as *tastes*. Tastes allow workers with the same skills to make different choices, whereas in Lazear's model skills alone determine who becomes a business owner. Tastes vary across workers with joint density $g(\tau_B, \tau_A)$, and τ_B and τ_A can be correlated because, for example, a worker who values autonomy is likely to value both types of business ownership. Tastes change how attractive an option is, but they do not change the returns to skills in that option. Because I take both skills and tastes as given and model their effect on the choice among W, B, and A, tastes do not influence how much of each skill a worker has in this framework.[9]

The worker chooses the option with the highest utility, where the utility of option j is its payoff plus her taste for it: $U_j = y_j + \tau_j$.[10] The utility gain from choosing option $j \in \{B, A\}$ over wage employment is $U_j - U_W = \tau_j - c_j$, where c_j denotes the opportunity cost of option

[8]The predictions in section 3.2 do not require additivity. They depend only on how the marginal return to each skill compares across options; for example, whether advanced business ownership pays more for BX than wage employment does ($r_A^{BX} > 1$). With additivity, the return to each skill does not depend on how much of the other skill the worker has, so these comparisons are the same for every worker. If, instead, the two skills are complements, the return to each skill depends on the level of the other skill, so a comparison such as whether advanced business ownership pays more for BX than wage employment does can be true for some workers and false for others, depending on how much SX each worker has. The predictions then apply to all workers as long as the comparisons are true over the range of skill levels in the data.

[9]In reality, tastes might affect how much of each skill a worker accumulates. For example, a worker with a strong preference for autonomy may deliberately choose a career in which she can accumulate more business skills, in anticipation of becoming a business owner. I discuss this in section 5.3.

[10]This form captures the trade-off between earnings and nonpecuniary benefits, such as being one's own boss. Studies of business ownership find that such benefits help explain why many workers become business owners despite lower earnings (Hamilton, 2000; Hurst and Pugsley, 2011).

j. Using equations 1–3, this opportunity cost can be expressed as

$$c_j = y_W - y_j = (1 - r_j^{BX})\, BX + (1 - r_j^{SX})\, SX, \qquad j \in \{B, A\}. \tag{4}$$

The worker prefers option j to wage employment when $\tau_j > c_j$.

Figure 1 shows the worker's choice for each possible pair of tastes (τ_B, τ_A). She chooses wage employment when neither taste exceeds its opportunity cost; that is, when $\tau_B < c_B$ and $\tau_A < c_A$. She chooses basic business ownership when $\tau_B > c_B$ *and* the gain from basic business ownership exceeds the gain from advanced business ownership ($\tau_B - c_B > \tau_A - c_A$), and she chooses advanced business ownership in the opposite case. The three regions meet at the point (c_B, c_A). Along the diagonal boundary $\tau_B - c_B = \tau_A - c_A$, the worker is indifferent between basic and advanced business ownership. Although the worker's choice is determined by her tastes (relative to opportunity costs), her skills determine the opportunity costs (c_B, c_A) and, in turn, the boundaries shown in figure 1.

3.2 Empirical Predictions

I use this framework to show how BX and SX are expected to affect the choice among wage employment, basic business ownership, and advanced business ownership, as well as payoffs after entry. In the model, a worker's choice is known once her skills and tastes are given. In the data, I observe her skills but not her tastes, so I treat her tastes as a draw from the density g and predict the probability of each choice. The probability that she chooses option j, P_j, is the probability that her tastes lie in the region of figure 1 that corresponds to option j.[11] Because skills enter the choice only through the opportunity costs, the effect of a skill on P_j works in two steps. The skill changes the opportunity costs (c_B, c_A), and the change in the opportunity costs then changes the probabilities.

In the first step, an increase in skill $k \in \{BX, SX\}$ changes the opportunity cost of option j by $\partial c_j / \partial k = 1 - r_j^k$, from equation 4. That is, skill k makes option j more attractive relative to wage employment when its return is higher in option j than in wage employment ($r_j^k > 1$). In the second step, a change in the opportunity costs shifts the boundaries in figure 1, which changes the probabilities. Consider an increase in c_B. The vertical boundary between wage employment and basic business ownership and the diagonal boundary both shift to the right. If her tastes lie just to the right of the vertical boundary, she now chooses wage employment, and if they lie just below the diagonal boundary, she now chooses advanced business ownership. As a result, the probability that she chooses basic business ownership falls, and the probability that she chooses advanced business ownership rises. An increase

[11] If each taste parameter is the sum of a constant and an independent, individual-specific type I extreme value term, the framework becomes the multinomial logit model of entry that I estimate in section 5.1. The coefficient on skill $k \in \{BX, SX\}$ in the log odds of option j relative to wage employment is then $r_j^k - 1$.

in c_A works the same way in the other direction. Hence

$$\frac{\partial P_B}{\partial c_B} < 0, \qquad \frac{\partial P_A}{\partial c_B} > 0, \qquad \frac{\partial P_B}{\partial c_A} > 0, \quad \text{and} \quad \frac{\partial P_A}{\partial c_A} < 0. \tag{5}$$

An increase in c_B or c_A increases the wage employment region in figure 1 and decreases the probability of becoming a business owner, regardless of the taste density g. By contrast, the signs of the effects of skills (BX and SX) on entry into and success in business ownership depend on the sizes of the returns. Below, I discuss the possible cases, argue which is most plausible, and state the associated prediction.

Business skills and entry into business ownership. Combining equations 4 and 5, the effect of BX on the probability of entering each type of business ownership is

$$\frac{\partial P_A}{\partial BX} = \frac{\partial P_A}{\partial c_A}(1 - r_A^{BX}) + \frac{\partial P_A}{\partial c_B}(1 - r_B^{BX}), \tag{6}$$

$$\frac{\partial P_B}{\partial BX} = \frac{\partial P_B}{\partial c_B}(1 - r_B^{BX}) + \frac{\partial P_B}{\partial c_A}(1 - r_A^{BX}). \tag{7}$$

If advanced business ownership pays more for BX than wage employment does ($r_A^{BX} > 1$), and basic business ownership pays less ($r_B^{BX} < 1$), both terms in equation 6 are positive and both terms in equation 7 are negative. I argue that this is the most plausible case. An owner of an advanced business uses her business skills to coordinate workers, allocate resources, and manage the business as a whole, allowing these skills to generate returns across the entire business (Lucas, 1978). By contrast, a wage worker applies her business skills within the scope of her assigned responsibilities, limiting the range of activities over which these skills generate returns. Basic business ownership involves simpler operations and less need for coordination and management, so the returns to business skills are likely to be relatively small. Other cases are also possible. If large firms offer managers higher returns to business skills than advanced business ownership does, then $r_A^{BX} < 1$, and BX increases the opportunity cost of both types of business ownership. However, as long as advanced business ownership pays more for BX than basic business ownership does ($r_A^{BX} > r_B^{BX}$), BX still shifts the diagonal boundary toward basic business ownership, raising the probability of advanced business ownership relative to basic business ownership. The most plausible case leads to my first testable prediction:

> **Prediction 1.** Business skills raise the probability of entering advanced business ownership and lower the probability of entering basic business ownership.

Specialized skills and entry into business ownership. An increase in SX changes the opportunity costs of basic and advanced business ownership by $1 - r_B^{SX}$ and $1 - r_A^{SX}$,

respectively. If both types of business ownership pay less for SX than wage employment does ($r_B^{SX} < 1$ and $r_A^{SX} < 1$), SX increases both opportunity costs and lowers the probability of becoming a business owner. I argue that this is the most plausible case. A business owner's expertise typically applies to only a subset of the tasks required to run the business, whereas a wage worker can devote more of her time to tasks that use her specialized skills. Of course, other cases are possible as well; for example, if a business relies directly on its owner's expertise, as in a professional practice or skilled trade, business ownership might offer higher returns to specialized skills than wage employment does. Even in the most plausible case, the framework does not determine the effect of a change in SX on the separate probabilities of entering basic and advanced business ownership. A rise in c_B leads some workers who would have chosen basic business ownership to choose advanced business ownership instead, and a rise in c_A does the reverse. Therefore, a rise in c_B can raise the probability of entering advanced business ownership even as the growing wage employment region lowers the probability of entering business ownership of either type. This leads to my second prediction:

> **Prediction 2.** Specialized skills lower the probability of entering business ownership of either type.

Skills and success in business ownership. Once the worker owns a business, the effect of a skill on her payoff is simply the return to that skill, r_j^{BX} or r_j^{SX} (equations 2 and 3). Both BX and SX raise payoffs in both types of business ownership as long as their returns are positive, which is invariably the case. For the reasons given for prediction 1, business skills are likely to raise payoffs more in advanced than in basic business ownership ($r_A^{BX} > r_B^{BX}$). The payoff effects also show the main difference between entry and success: Entry depends on how a return compares with wage employment ($r_j^k - 1$), whereas success depends on the return itself (r_j^k). As a result, a skill with $0 < r_j^k < 1$ can lower entry into business ownership and still improve success once the worker owns a business. This cannot happen in Lazear's model, where the skill that raises a business owner's payoff, her weaker skill, also makes entry more likely. Given these arguments, my final prediction is:

> **Prediction 3.** Business skills and specialized skills both raise payoffs in business ownership, and business skills raise payoffs more in advanced than in basic business ownership.

4 Data

4.1 Data Sources

In conducting my empirical analysis, I combine data from the 1979 National Longitudinal Survey of Youth (NLSY79) with data from O*NET (described below). The NLSY79 began in 1979 with a nationally representative sample of 12,686 individuals born between 1957 and 1964 and residing in the United States. The respondents were surveyed annually from 1979 to 1994 and biennially thereafter. I use survey waves from 1979 through 2020.

The NLSY79 is uniquely well-suited for my study for a number of reasons. First, it follows respondents over a long horizon. I am able to track respondents' work histories from school exit until their last interview date, which in many cases affords a four-decade observation window. This long coverage is particularly valuable for studying business ownership, which is often entered in mid- or late career. Second, the survey tracks employment, occupation, and nonemployment spells in detail. This enables me to construct precise measures of the experience and skills accumulated at any point in time and to observe transitions into and out of each job, where a job is a spell with one employer. Third, the survey identifies which jobs are self-employment or self-owned businesses. In addition to the traditional class-of-worker variable, which classifies each job as self-employment or wage employment, a separate business ownership module which was introduced in 2010 identifies all businesses that respondents own (retrospectively to age 18 and contemporaneously from 2010 onward) and links them to specific job spells. Fourth, it provides ample detail on jobs (such as earnings and hours worked), businesses (such as revenue, incorporation status, and start-up capital), and individuals (such as educational attainment, cognitive and noncognitive abilities, family and household structure, and wealth).

I also use data from version 18.1 of the Occupational Information Network (O*NET), which is administered by the U.S. Department of Labor.[12] The O*NET database contains a wealth of occupation-specific information. I use the so-called "level" and "importance" scores, which indicate both the amount of and importance of dozens of skill and knowledge categories in each of 873 occupations. By linking these occupation-specific scores to each NLSY79 sample member's occupations, I am able to construct my time-varying measures of business skills, specialized skills, and skill balance.

[12]This information is obtained on an ongoing basis, starting in 1998, via surveys of randomly sampled workers and from input provided by occupational experts. As a result, the data represent occupational conditions over the last three decades only, and are not year-specific.

4.2 Sample Construction

As discussed in detail in section 5, I estimate three models: a competing-risks model for the decision to transition into business ownership, and both a business revenue model and an earnings model for workers who are business owners. I construct a separate sample of NLSY79 respondents for each model.

Sample for the entry model. To model the year-by-year decision of individuals who are not business owners (that is, wage workers and the nonemployed) to enter either basic or advanced business ownership, I define the career start date for each respondent as the month and year in which the first nonenrollment spell lasting at least 12 months begins. Starting with the full sample of 12,686 NLSY79 respondents, I first exclude 392 individuals who leave the survey before their career start date or before holding at least one job after the start date. Second, I exclude 936 individuals with incomplete work histories after the career start date (for example, missing job start or stop dates). Third, I exclude 4,037 individuals who exit the survey before 2010, when the business ownership module was added. Fourth, I exclude 25 individuals who are only business owners (without any wage employment) throughout their observed careers. Finally, I exclude 180 individuals with left-censored careers, meaning they started their careers before the survey began tracking work experiences. These restrictions leave a sample of 7,116 individuals.

Each of these 7,116 sample members contributes one person-year observation to the sample for every 12-month period that they are not business owners. This sequence of observations begins at the career start date, and ends when they enter business ownership or are last interviewed. However, individuals who leave business ownership reenter the sample and contribute additional observations. The final sample used for the entry model contains 251,214 person-year observations for 7,116 individuals.

Samples for the business revenue and earnings models. To model business revenue, I return to the sample of 7,116 individuals described above and restrict it to those who ever experience business ownership. Each of these individuals contributes one observation for each year for which they are business owners and business revenue is reported. For businesses without a yearly report, such as those that ended before 2010, I use the revenue that respondents report retrospectively for a typical year. A limitation of NLSY79 data is that business revenue is collected only from 2010 onward, and only for jobs identified

by respondents as self-owned businesses.[13] Because of this limitation, the resulting sample contains only 2,430 observations for 938 individuals with 1,221 basic businesses, and 2,465 observations for 832 individuals with 1,027 advanced businesses. As discussed below, I use the more broadly available earnings data to assess how these sample restrictions affect the results.

To model earnings among individuals who are currently business owners (and/or self-employed, per footnote 13), I use a sample selection strategy similar to the one I use for the revenue sample. Starting with the same 7,116 individuals, I allow each individual to contribute one observation for each business–year for which weekly earnings are available. This produces a sample of 10,383 observations for 2,370 individuals with 4,118 basic businesses, and 5,938 observations for 1,103 individuals with 1,554 advanced businesses. I also estimate the earnings model on the subset of these observations for which revenue is also available.

4.3 Business Types and Outcome Variables

In modeling entry into business ownership, business revenue, and earnings among business owners, I distinguish between two types of business ownership (basic and advanced) using a combination of incorporation status and start-up capital at entry.[14] Many studies separate business owners by incorporation status (Levine and Rubinstein, 2017, 2018; Humphries, 2017; Can, 2022; Can and Fossen, 2022). Incorporation status is an important dimension because it captures the owner's legal commitment to the business. However, the decision to incorporate can also reflect tax or liability considerations, and businesses with the same legal form can differ widely in how much they invest at entry. On average, incorporated businesses start with more capital. As shown in figure 2, among incorporated businesses with positive start-up capital the median is $12,427 and the mean is $111,309, compared with $3,883 and $27,143 for unincorporated businesses. Figure 2 also shows that many incorporated businesses start with little capital, while some unincorporated businesses start

[13] From 1979 onward, respondents report the class of worker for every job held, which identifies a subset of jobs as self-employment. Starting in 2010, respondents retrospectively report every business owned since age 18, and they report new businesses in each later interview. I link each business to the corresponding job in the NLSY79 work history, which records the start and end dates and the occupation of every job a respondent has held. I drop the few businesses that cannot be linked to any job. Many jobs are classified both as self-employment and as a business, but some are classified as only one of the two (Light and Munk, 2018). Owners of incorporated businesses might report themselves as employees of their own firms, while workers in small, home-based businesses might not report their work as a business. I use the universe of self-employment jobs and self-owned businesses to model entry into business ownership and, as described below, to model earnings, but I can only use jobs classified as businesses (many of which are also classified as self-employment) in the revenue analysis.

[14] Another common approach to classifying business ownership is to consider firm size, typically measured by the number of employees. Firm size was not collected from 1981 to 1985 in the NLSY79, which might lead to inconsistent measurement across career stages.

with $50,000 or more. Because start-up capital captures a dimension of business ownership that incorporation status alone does not (financial resources committed at entry), I use both measures to classify businesses.

I classify a business as basic if it is unincorporated *and* its start-up capital is less than $5,000 in 2010 dollars.[15] I classify a business as advanced if it is incorporated *or* its start-up capital is at least $5,000. The cutoff of $5,000 is approximately the median value among all businesses with positive start-up capital.

Unsurprisingly, basic business ownership is far more common than advanced business ownership. Of the 4,569 businesses transitioned into in the entry sample, 76% are basic and 24% are advanced. As indicated in section 4.2, basic businesses account for 73% of the businesses used to model earnings, but only 54% of those used to model business revenue, given the restrictions on that sample.

When modeling entry into business ownership, I use the classification just described to determine whether, on a year-by-year basis, a worker who is not currently a business owner remains in her current state, becomes a basic business owner, or becomes an advanced business owner. Because I define entry as a transition from not owning a business to owning one, I do not count a direct move from one business to another as an entry.

When modeling "success" outcomes among business owners, I use both business revenue and earnings. As noted in section 4.2, business revenue is collected only from 2010 onward and only for jobs that respondents identify as businesses, while earnings are reported for every job from 1979 onward. At each interview, respondents report their current or most recent "rate of pay" for every job held since the prior interview. Some report this as weekly earnings. In instances where pay is reported as hourly, biweekly, monthly, or in other units, I use reported weekly hours to construct a measure of weekly earnings. I use weekly rather than hourly earnings because business owners can often adjust their hours more flexibly than can other workers, so differences in earnings may reflect both hourly returns and hours worked. In estimating the revenue and earnings models, I use a log transformation of each dependent variable, after assigning one dollar to anyone reporting zero earnings.

4.4 Independent Variables

I first discuss the three time-varying, O*NET-based skill measures—business skills, specialized skills, and skill balance—which are my key independent variables. For each worker, I compute all three variables for every year from the career start date onward, using her occupational history up to that year. The entry model uses their values in each year in which the

[15] Throughout this analysis, I use the annual Consumer Price Index for four geographic regions (Northeast, Midwest, South, and West) to express all dollar amounts in 2010 dollars.

worker is at risk of entry, and the revenue and earnings models use their values at the start of each business. I then explain how I construct my occupation count variable, which is the conventional measure of skill balance, and summarize the set of baseline regressors included in each model.

4.4.1 Business Skills

To construct my measure of business skills, BX_{it} (which is the empirical counterpart of BX in section 3), I combine O*NET data on the business-skill intensity of each occupation with workers' occupational histories. I proceed in three steps. First, I identify the O*NET skill and knowledge elements that capture business skills. Second, I use these elements to measure the business-skill intensity of each occupation. Third, I combine these intensities with the experience worker i has accumulated in each occupation from the career start date to time t to construct BX_{it}.

In step 1, I identify those O*NET skill and knowledge elements that capture three components of establishing and operating a business: core business knowledge, managerial skills, and interpersonal skills. Core business knowledge includes economics and accounting, sales and marketing, production and processing, customer and personal service, and service orientation. Managerial skills consist of administration and management, personnel and human resources, and the management of financial, material, and personnel resources. Interpersonal skills consist of social perceptiveness, coordination, persuasion, negotiation, and instructing; the first four are the items that Deming (2017) uses to measure social skills. In making this selection, I use the O*NET elements that rank highest for chief executives (CEOs) as a reference, because the CEO occupation encompasses key activities of business ownership, including making financial and strategic decisions, managing staff and resources, and interacting with customers or clients. This process identifies 15 of O*NET's 68 total skill and knowledge elements as what I refer to as "business elements." Table A1 lists all 68 elements scored by O*NET and identifies the 15 that I select to construct the business skill measure.

In step 2 of the construction process, I use these 15 business elements to assign each occupation a business-skill intensity. O*NET rates each of its 68 skill and knowledge elements in each of its 873 occupations on two dimensions: a level score from 0 to 7, measuring how much of the element the occupation requires, and an importance score from 1 to 5, measuring how central the element is to the occupation. For my 15 chosen business elements, I rescale both scores to range from 0 to 1 and refer to their product as the element score.[16] I

[16]The element score takes into account both how much of an element the occupation requires and how central the element is to the work. In practice, the two scores are highly correlated. Across the occupations held by workers in my sample, the correlation between level and importance is 0.96, and each has a correlation of 0.94 with their product.

then confine my attention to element scores for the 587 occupations held by NLSY79 respondents, calculate the average element score among the 15 business elements for each of these occupations, and normalize each average by the highest value among all 587 occupations.[17] This normalization sets the most business-intensive occupation (CEO) to 1 and expresses the intensity of every other occupation relative to that benchmark.

In step 3, I combine these occupational intensities with workers' occupational histories to construct BX_{it}. For worker i at time t, I define

$$BX_{it} = \sum_{o \in O_i} \omega_o^{BX} \cdot OccExp_{iot}, \tag{8}$$

where O_i is the set of occupations worker i has ever held and $OccExp_{iot}$ is her cumulative experience in occupation o from the career start date to t, measured in years. The weight ω_o^{BX} is the normalized business-skill intensity of occupation o described above.

For example, 10 years of uninterrupted employment as a CEO (the most business-intensive occupation with an intensity of 1) yield a BX_{it} of 10. Ten years as a purchasing manager, whose intensity is 0.81, yield a BX_{it} of 8.1, while 10 years as an agricultural product grader or sorter, one of the least business-intensive occupations with an intensity of 0.08, yield a BX_{it} of 0.80. Table 1 reports means and standard deviations of BX_{it} (and all other covariates) for each sample used in estimation.

A potential concern is that BX_{it} captures general skill requirements rather than business-specific skills, because occupations that require and use more business skills are likely to also require more education, training, and prior experience. To assess this concern, I turn to O*NET job zone classifications, which score each occupation on the preparation it requires, ranging from zone 1 (little or no preparation) to zone 5 (extensive preparation); zone 4–5 occupations generally require a bachelor's degree. I construct a measure of each worker's years of experience in zones 4–5 ("Actual experience in Job Zone 4+ occupations"), and include this proxy for general skill requirements in my business ownership entry model with and without my key regressors. As reported in table A2, when my key regressors are excluded (column 2), the general skill requirements measure is associated with lower entry into basic business ownership (the estimated marginal effect is -7.1%) and higher entry into advanced business ownership (the estimated marginal effect is 9.0%). However, when BX_{it} and other key regressors are included (column 3), the estimated effects of "Actual experience in Job Zone 4+ occupations" are much closer to zero and imprecisely estimated. Moreover, the

[17] The NLSY79 codes occupations with the 1970 Census classification through 2000 and the 2000 Census classification thereafter, whereas O*NET uses the 2010 Standard Occupational Classification (SOC). I link both NLSY79 classifications to the 2010 SOC using the crosswalks developed by Lise and Postel-Vinay (2020).

estimated marginal effects for BX_{it} are largely invariant to the inclusion or exclusion of this general skill requirements proxy (columns 1 vs. 3). These findings indicate that BX_{it} captures skills specific to establishing and operating a business rather than general skill requirements.

4.4.2 Specialized Skills

My specialized skills variable, SX_{it}, measures the extent to which a worker's experience up to time t matches the specific expertise required by her target business. For example, a worker who starts a law firm needs expertise in law and government, whereas a worker who starts a plumbing business needs expertise in plumbing/mechanics and building/construction. Because the expertise required varies across businesses, the measure is tailored to each worker's target occupation. I define the *target occupation* as the occupation held at the start of the business for workers who enter business ownership, and as the occupation held 18 years into the career for those who never own a business, where 18 corresponds to the average experience at entry into business ownership in the sample.[18]

To determine which specialized skills are required in each target occupation, I rank all 68 O*NET skill and knowledge elements (see table A1) by their element scores, defined in section 4.4.1 as the product of the rescaled importance and level scores. I select the top one to three elements, using natural breaks in the ranking, to identify elements that stand out from the rest in each occupation. I then exclude elements that rank near the top across most occupations, because broadly required skills do not distinguish a worker's area of specialization. For example, the top three elements for lawyers are law and government, English language, and speaking. Because English language and speaking rank near the top in many occupations, I keep only law and government as the specialization element. For plumbers, the top two elements are mechanical and building/construction, and both are kept.[19]

I construct specialized skills using the same approach as the business skills measure, but with weights tailored to each worker's target occupation. For worker i at time t, I define

$$SX_{it} = \sum_{o \in O_i} \omega_{io}^{SX} \cdot OccExp_{iot}, \tag{9}$$

where the weight ω_{io}^{SX} is the average, across worker i's specialization elements, of occupation

[18] Alternatively, I could define the target occupation for workers who do not enter business ownership as the occupation in which they have spent the most time. The estimates are robust to which measure I use.

[19] This procedure is similar in spirit to TF-IDF (term frequency–inverse document frequency) weighting in natural language processing, where element scores capture the strength of an element within an occupation, while excluding elements that rank near the top across most occupations serves to remove elements that are not distinctive to a particular occupation.

o's element score divided by that element's highest score among the occupations held by sample members. An occupation receives a weight of 1 if no other occupation uses any of worker i's specialization elements more intensively. Because the weights depend on the target occupation, two workers with identical work histories can have different values of SX_{it} if their target occupations differ. As a result, the measure captures experience in occupations that develop the specific expertise required by a worker's target business, rather than work experience in general.

4.4.3 Skill Balance

My direct measure of skill balance, which I call *balanced skill count* (BSC_{it}), captures how evenly a worker's accumulated skills are distributed across O*NET's 68 skill and knowledge elements. I construct it in three steps. First, I divide each element score by its highest value across the occupations held by workers in my sample, so that all elements are measured on the same scale. Second, for each element, I compute the worker's accumulated skill by summing these rescaled scores across all occupations she has held, weighted by the years she spent in each; then I keep the 30 elements in which she has accumulated the most skill.[20] Third, letting p_{its} denote the share of element s in her accumulated skill across these 30 elements, I summarize their distribution using the exponentiated Shannon index

$$BSC_{it} = \exp\left(-\sum_{s} p_{its} \log p_{its}\right). \tag{10}$$

The variable BSC_{it} ranges from 1, when all accumulated skill is concentrated in a single element, to 30, when it is distributed equally across all 30 elements. A higher value indicates a more balanced skill portfolio, while a lower value indicates greater concentration in a few elements.[21]

4.4.4 Number of Occupations and Involuntary Job Losses

To compare my direct measure of skill balance, BSC_{it}, with the approach used in prior work, I also use the occupation count, which is one of the most common proxies for skill balance. The occupation count is the number of distinct occupations a worker has held from

[20] Workers typically accumulate little or no skill in many elements, so including all 68 would mechanically make every worker's skills appear unbalanced, even when they are evenly distributed across the elements relevant to her work experience. On average, the top 30 elements account for about 80% of a worker's accumulated skill across all 68 elements.

[21] A common alternative is the inverse Herfindahl–Hirschman index (HHI), $1/\sum_s p_{its}^2$. Because the HHI squares skill shares, it is more sensitive to changes in larger shares and less sensitive to changes in smaller shares. In contrast, the exponentiated Shannon index places relatively more weight on smaller shares, making it better suited to Lazear's model, in which a business owner's payoff depends on her weakest skill.

the career start date to time t. I count occupations using the 23 two-digit occupations in the 2010 Standard Occupational Classification (SOC). The count increases when a worker enters a two-digit occupation she has not previously worked in, but a return to a previously held occupation does not increase the count. Unlike business skills, specialized skills, and the balanced skill count, this measure does not use O*NET data.

To illustrate how my direct measure of skill balance (BSC_{it}) compares to the conventional occupation count, consider two workers, each with 15 years of experience divided equally across three occupations. The first worker has worked as a bank loan officer, a debt collector, and a salesperson. Because all three occupations draw heavily on the same elements, such as dealing with customers, persuading, and negotiating, her accumulated skill portfolio is concentrated in a relatively small set of elements and her value of BSC_{it} is 26.6. The second worker has worked as a salesperson, a truck driver, and a carpenter. These occupations draw on more distinct elements, spanning sales, vehicle operation, and building/construction. As a result, her accumulated skill portfolio is more evenly distributed across elements and her BSC_{it} is 29.5. In short, these workers have very different values of BSC_{it} (near the bottom and the top of the sample distribution, respectively), even though both have an occupation count of three.

I also construct a time-varying count of involuntary job losses to assess whether the occupation count captures job churning rather than skill balance. For each job that ends, the NLSY79 records the reason for separation. I classify a job ending as an involuntary job loss if the worker was laid off, discharged, or fired, if the plant closed, or if a temporary job, seasonal job, or program ended. The count is the number of involuntary job losses from the career start date to time t.

4.4.5 Baseline Regressors

In my empirical analysis, I include a rich set of control variables. The choice of controls follows a large literature on the determinants of business ownership, including risk tolerance (Ahn, 2010; Hsieh, Parker and van Praag, 2017), access to credit and household wealth (Hurst and Lusardi, 2004; Fairlie and Krashinsky, 2012), and family background (Hout and Rosen, 2000; Hvide and Oyer, 2025). As noted earlier, table 1 provides means and standard deviations for all covariates for all samples; it also provides a clear overview of which regressors are included and excluded in each model.

I control for past work experience in two ways. Potential experience is the time elapsed since the career start date, and actual experience is the time actually worked, adjusted for the average weekly hours worked on each job. Both are measured in years and enter along with their squares. I also control for the number of past business ownership spells and (for

the revenue and earnings models only) years of tenure in the current business.

Individual characteristics include dummy variables identifying gender, race/ethnicity (Hispanic, Black, or neither), marital status (never married, currently married, or previously married), whether the respondent is cohabiting with an unmarried partner, and whether the youngest child in the household is age six or younger.[22] I also control for education (less than high school, high school graduate, some college, or college or above) and the Armed Forces Qualification Test (AFQT) percentile score as a measure of cognitive ability. As measures of noncognitive traits, I include risk tolerance (four categories based on the NLSY79's income-gamble questions), sociability at age six (four categories from extremely shy to extremely outgoing), the Rotter Locus of Control score, and two of the Big Five personality traits (extraversion and openness to experience) measured with self-rated items from the Ten-Item Personality Inventory (TIPI). I also use a binary indicator of whether the respondent considers herself an entrepreneur as a proxy for her taste for business ownership. Because this latter measure is available only from 2010 onward and may reflect prior business ownership experience, I use it only in the sensitivity analysis discussed in section 5.3.

For family background, I control for net family wealth, net family income, whether the respondent's father had at least some college education, whether the father worked in a professional or managerial occupation in 1979, and whether any family member owned a business. Because net family wealth is available only from 1985 onward, I impute it for earlier years. I exclude net family wealth and income from the revenue and earnings models because both measures include business income for business owners and thus partly capture the outcomes of interest.

The last regressors shown in table 1 are two sets of dummy variables indicating geographic region and industry of employment, both of which are included in the business revenue and earnings models but not the business ownership entry model. To reiterate, the absence of summary statistics from a given column of table 1 indicates that the variable is not included in the model for which the given sample is used. Table 1 also indicates which regressors are time-varying.

4.5 Descriptive Statistics

Before turning to my empirical strategy and findings, I establish some basic patterns in the data. The descriptive statistics in this subsection (tables 2–6) show (a) how many individuals become business owners and how many enter business ownership more than once; (b) which occupations are most common in each type of business ownership; (c) how entrants into

[22]Having a young child can motivate entry into self-employment as a means to balance work and family, especially for women. See Lim (2019) and Lim and Michelmore (2018).

basic and advanced business ownership differ from each other; (d) how basic and advanced businesses differ in their characteristics; and (e) how strongly my skill and count measures are correlated with each other.

Table 2 shows how often the 7,116 individuals in the entry sample enter business ownership (panel A) and which types of businesses they enter overall (panel B). Both panels reveal that, over the sample period, 4,166 individuals (58.5%) never own a business and the remaining 2,950 individuals (41.5%) enter business ownership at least once. This is a high rate of business ownership, but it reflects the fact that the sample follows individuals for nearly 40 years and includes business ownership in both main and side jobs. For comparison, in July 2010, when sample members are aged 45–53, 13.4% of employed individuals in the sample own a business as their main job, which is very close to the 13.5% of workers aged 45–54 who are self-employed in the 2009 Current Population Survey (Hipple, 2010). Including side businesses raises the share in the sample to 17.0%.

Panel A of table 2 reveals that, among individuals who ever enter business ownership, 1,838 (25.8% of the full sample and 62.3% of business owners) do so only once, while another 761 (10.7% of the full sample and 25.8% of business owners) own two businesses and the remaining 4.9% of the full sample own between three and eight businesses over their observed careers. In panel B we see, as reported in section 4.3, that basic businesses are far more common than advanced businesses: Among all 7,116 individuals in the sample, 28.0% enter only basic businesses, compared to 8.4% who enter only advanced businesses and 5.1% who enter both types at different points in their careers. These patterns show that business ownership—and especially basic business ownership—is quite common over the course of workers' long careers, and that multiple business ownership is largely limited to two businesses. In light of these patterns, I choose to include every business entry (and not only initial entries) in my business ownership entry model and to allow the same individual to enter different types of businesses at different times.

Table 3 provides information on the kinds of businesses these individuals run by listing the eight most common occupations in each type of business. Common basic businesses include personal services and manual trades such as childcare, landscaping, cleaning, painting, and construction labor. Advanced businesses are more concentrated in managerial occupations, such as managers, chief executives, retail sales supervisors, and food service managers. A few occupations, such as sales workers, carpenters, and managers, appear in both lists, so occupation alone clearly does not determine whether the business is incorporated and/or relatively capital-intensive.

In table 4, I follow up on the summary statistics in table 1 to show how select characteristics differ between basic and advanced business owners; here, the unit of observation is

the last person-year observation (from the business entry sample) before the individual transitions into the given type of business ownership. The first row reveals that basic entrants tend to start their businesses earlier than do advanced entrants: on average, basic entrants are 35 years old and have 10.9 years of actual experience when they enter, whereas advanced entrants are 38.5 years old and have 15.3 years of actual experience. Advanced entrants also accumulate more business skills than basic entrants prior to entry: their average BX_{it} is 6.0 compared to 3.6 for basic entrants. Because BX_{it} grows with time in the labor market, part of this gap reflects the later entry of advanced entrants. However, a small gap remains (0.38 vs. 0.31) when business skills are measured per year of actual experience; clearly, the careers of advanced business owners are concentrated slightly more heavily in business-intensive occupations prior to entry. In contrast, specialized skills and the balanced skill count differ little between the two types when adjusted for years of experience. Table 4 reveals that the two types of business owners also differ in their history of job churning. Basic entrants hold 0.71 occupations per year of actual experience, on average, versus only 0.48 for advanced entrants, and experience almost twice as many involuntary job losses per year, on average (0.32 vs. 0.17). Finally, advanced entrants tend to be more educated and score higher on the AFQT than basic entrants; for example, 31% of advanced business owners have a college education, compared to only 18% of basic business owners.

In table 5, I switch from comparing individual characteristics to comparing characteristics of basic and advanced businesses. Unsurprisingly, advanced businesses tend to generate more revenue than basic businesses, provide higher earnings to their owners, and last longer: excluding top-coded values, average annual business revenue is $215,000 for advanced businesses and only $63,000 for basic businesses, and revenue is top-coded for 9% of advanced businesses but only 1% of basic businesses. Owners of advanced businesses earn $844 per week on average, which is almost twice the $443 average earnings among owners of basic businesses; they also work about nine more hours per week, on average. Advanced businesses last 7.6 years, on average, compared to 3.9 years for basic businesses, and are more likely to last five years or more (49% vs. 23%). Because advanced businesses tend to last longer than basic businesses, they are also much more likely (32% vs. 12%) to be right-censored (that is, ongoing at the last interview).

Because my skill and count measures enter the models together, table 6 reports how strongly they are correlated with one another. All five measures grow with time in the labor market, so the table reports partial correlations that control for actual experience. Panel A of table 6 uses the person-years in the entry sample, and panels B and C use basic and advanced businesses in the earnings sample, measured at the business's start. The patterns are similar across the three panels: correlations between BX_{it} and SX_{it} are moderate, ranging from 0.40

to 0.56, partly because in some occupations, such as sales, the specialization elements are business elements. The two measures are, therefore, related but distinct. The correlations between BSC_{it} and BX_{it} are weak (between 0.15 and 0.21) and between BSC_{it} and SX_{it} are close to zero, so skill balance is clearly largely independent of how much business or specialized skill a worker has accumulated. Occupation count follows a different pattern: its correlations with BSC_{it} range from 0.18 to 0.25, its correlations with involuntary job losses range from 0.36 to 0.48, and its correlations with BX_{it} and SX_{it} are negative. This suggests that occupation count (a common proxy for skill balance in tests of the JOAT hypothesis) partly reflects job churning rather than skill balance alone. Taken together, these correlations indicate that BX_{it}, SX_{it}, and BSC_{it} capture distinct aspects of work experience, each with enough independent variation to be included in the same model.

5 Empirical Strategy

5.1 Entry into Business Ownership

I use a sequential, discrete-time, competing-risks model for the probability of entry into business ownership. In each year, a worker who is not currently a business owner faces three options. She can enter basic business ownership, enter advanced business ownership, or stay out of business ownership. The latent value of option j for worker i in year t is

$$V_{ijt} = \alpha_{0j} + \alpha_{1j}BX_{it} + \alpha_{2j}BX_{it}^2 + \alpha_{3j}SX_{it} + \alpha_{4j}SX_{it}^2 + \alpha_{5j}BSC_{it} + \alpha_{6j}BSC_{it}^2 + X_{it}\alpha_{Xj} + \varepsilon_{ijt}, \tag{11}$$

where BX_{it} is business skills, SX_{it} is specialized skills, BSC_{it} is the balanced skill count, and X_{it} is the set of baseline regressors described in section 4.4.5 and table 1. The worker chooses the option with the highest value. The coefficients depend on j because each covariate can affect entry into basic and advanced business ownership differently. Each skill measure enters with a quadratic term because the return to each skill measure might change as it grows. Besides this main specification, I estimate versions in which BX_{it}, SX_{it}, and BSC_{it} enter individually and jointly, and versions that include the number of occupations held to date and the number of involuntary job losses. I assume that the errors ε_{ijt} are independent across options and follow a type I extreme value distribution, which yields a multinomial logit (MNL) model with the coefficients for staying out of business ownership normalized to zero. Standard errors are clustered at the individual level to allow for correlation in the errors over time for a given individual. For comparison, I also estimate a binomial logit that pools entry into basic and advanced business ownership.

5.2 Business Revenue and Earnings Models

As noted earlier, I model both business revenue and earnings among business owners as two alternative measures of business success. Revenue measures the scale of the business, but it does not subtract costs, and the data limitations described in section 4.2 leave me with a restricted sample. Weekly earnings are available for nearly every business and often for multiple years for a given business, but they understate the owner's payoff when owners reinvest profits or pay themselves little for tax reasons. Because neither outcome measures the payoff y_j envisioned in the conceptual framework (section 3) exactly, I examine both. I estimate both the revenue model and the earnings model separately for observations associated with basic and advanced businesses, because the conceptual framework predicts that business skills and specialized skills are rewarded differently depending on the type of business.

For worker i in year t in a business of type $j \in \{B, A\}$, the model is

$$\ln y_{it} = \beta_{0j} + \beta_{1j} BX_{it} + \beta_{2j} BX_{it}^2 + \beta_{3j} SX_{it} + \beta_{4j} SX_{it}^2 + \beta_{5j} BSC_{it} + \beta_{6j} BSC_{it}^2 + W_{it}\beta_{Wj} + u_{it}, \tag{12}$$

where y_{it} is either annual business revenue or weekly earnings, and W_{it} contains the control variables described in section 4.4.5 and table 1. The relationships between BX_{it} and SX_{it} and the outcome—which, here, are allowed to be quadratic—correspond to the skill-specific returns r_B^{BX}, r_B^{SX}, r_A^{BX}, and r_A^{SX} in the conceptual framework.

When the outcome in equation 12 is log annual revenue, I estimate the model using a Tobit because revenue is top-coded in the NLSY79. When the outcome is log weekly earnings, I estimate the model by OLS. As discussed in section 4.2, I estimate the earnings model with two samples: the first is the restricted sample of observations for which revenue is also available, and the second is the full earnings sample. A comparison of estimates for the two earnings samples indicates whether my restricted revenue sample is nonrepresentative. Standard errors are clustered at the individual level in all models.

5.3 Selection on Observables

Workers with a strong taste for autonomy, to consider one example, are likely to accumulate relatively large amounts of BX_{it} because they hope to own a business. If I cannot control for this taste, I cannot separate the direct (causal) effect of BX_{it} on entry into business ownership from the confounding effect of the taste for autonomy. The same problem arises in the revenue and earnings models, where unobserved characteristics may be related to both the skills workers accumulate and how well their businesses perform. This is a classic endogeneity problem, although in my application it is not essential that a causal effect of BX_{it} (and other key regressors) be identified because I am not attempting to assess the

value of, say, a government program designed to provide BX_{it} to a population of workers who differ widely in their unobserved taste for autonomy. More generally, the JOAT literature and the broader labor economics literature that seeks to identify relationships between work experience-related measures and labor market outcomes do not typically address the endogeneity of work experience.[23]

Although a strictly causal estimate of the effect of BX_{it} (and other key regressors) on each outcome is not essential, my estimation strategy does not ignore endogeneity insofar as it relies on selection on observables. That is, conditional on my extensive set of baseline regressors, I assume that my key skill regressors are largely independent of unobserved factors that affect each outcome. This assumption cannot be tested directly, but when presenting my findings in section 6, I consider how the estimated marginal effects of the skill measures in the entry model change as I vary the set of controls, focusing on those controls most closely related to tastes for business ownership: personality traits (including risk tolerance), whether a family member owned a business, and whether the respondent considers herself an entrepreneur. If *unobserved* tastes are related to the skill measures in the same way as these *observed* taste proxies are, then relatively small changes in the estimates when these proxies are added suggest that controlling for unobserved tastes would also have little effect on the estimates. In contrast, a lack of robustness suggests that unobserved tastes may play a more important role (Altonji, Elder and Taber, 2005).

In appendix B, I also consider two alternatives to selection on observables for addressing endogeneity issues. The first is individual fixed effects, which absorb time-invariant tastes. They are of limited used in my entry model because little within-person variation in skills remains once experience is controlled for, but I use them to compare the returns to skills in business ownership and wage employment for the same workers. The second is a pair of instruments based on the industrial composition of the county where a worker starts her career, but these instruments are too highly correlated to separate business skills from specialized skills. I hope to pursue alternative estimation methods in future work.

As noted at the outset of this subsection, the lack of strictly causal estimates of the effects of my key regressors (especially BX_{it} and SX_{it}) is not as concerning as it would be in many other applications. In particular, the comparison of my findings with conventional tests of the JOAT hypothesis (most of which also rely on selection on observables) focuses on which dimensions of work experience predict entry into business ownership, rather than on identifying the causal effects of each skill measure. I am interested in learning what these

[23] A notable exception is the identification of the relationship between job tenure and wages, where causality *is* a primary concern because of the assumed relationship between tenure and unobserved "match quality" (Topel, 1991).

associations reflect (for example, whether the relationship between occupation count and business entry reflects skill balance or job churning) and not in providing a policy-relevant analysis of causality. In addition, the most likely source of endogeneity is that unobserved tastes for business ownership are positively correlated with BX_{it}, leading to an upward bias in the estimated effect of BX_{it}. This bias cannot account for my finding (which is consistent with my theoretical predictions) that business skills are predicted to promote entry into advanced business ownership but detract from entry into basic business ownership.

5.4 Average Marginal Effects

To draw inferences from the estimated coefficients of each model, I compute average marginal effects. For each observation in the sample, I use the MNL, Tobit, or OLS estimates to compute the change in the predicted outcome associated with a one-unit increment in each skill measure, holding all other variables at their observed values, and then average these changes over the sample. In the entry models, I express these effects as percentage changes in the predicted probability of each type of entry. In the revenue and earnings models, the outcomes are in logs, so the average marginal effects are changes in log points, which I describe as approximate percentage chages. Average marginal effects facilitate the interpretation of the key relationships across specifications and models, given that I use a nonlinear model for entry and allow each skill measure to have a nonlinear relationship with each outcome. I standardize BX_{it}, SX_{it}, and BSC_{it} prior to estimation, so that a one-unit increment corresponds to a half-standard-deviation increase. This is approximately two years as a CEO for business skills and two and a half years for specialized skills in the occupation that uses a worker's specialization elements most intensively.

6 Findings

I first discuss the estimates of the competing-risks multinomial logit (MNL) model and binomial logit model of entry into business ownership (section 6.1) and then turn to estimates of the revenue and earnings models for business owners (section 6.2). The discussion focuses on the estimated marginal effects of business skills, specialized skills, and skill balance on each outcome, and on whether the estimates support my three innovations to the JOAT literature. Results of comparing my direct measure of skill balance with occupation count are discussed for the entry model only. As discussed in section 5.4, I focus on average marginal effects, which are reported in tables 7–9 and computed from the coefficient estimates in appendix tables A3, A4, and A5. Each average marginal effect is the change in the predicted outcome from a half-standard-deviation increase in the given measure.

6.1 Entry into Business Ownership

Before turning to the estimates for my MNL and binomial logit models of entry into business ownership, I restate the arguments they are designed to test. First, I argue that business skills and specialized skills—and *not* a balance across all skills—are rewarded in business ownership. Based on what I judge to be the most plausible assumptions about relative returns to these skills in wage employment, basic business ownership, and advanced business ownership, I expect business skills to *increase* the probability of entry into advanced business ownership but *lower* the probability of entry into basic business ownership (prediction 1), and I expect specialized skills (which are most highly valued in wage employment) to lower the probability of entering business ownership of either type (prediction 2). Moreover, I expect that my direct measure of skill balance will not predict entry into business ownership once business skills and specialized skills are controlled for. Second, because business skills are predicted to have opposite effects on entry into basic and advanced business ownership, it is readily apparent that a (binomial logit) model that fails to distinguish between the two types will obscure these patterns. Third, I argue that existing evidence that skill balance *is* an important determinant of business ownership based on "occupation count" (or employer count) as a crude proxy for skill balance is misleading because the number of occupations (or employers) encountered to date might reflect an unproductive form of "job churning" rather than skill balance.

In panel A of table 7, I report estimated average marginal effects of these key covariates on the probabilities of entering basic and advanced business ownership, based on MNL estimates reported in appendix table A3; each column of table 7 corresponds to a different specification of the MNL model. Focusing first on column 5, which corresponds to equation 11 (section 5.1) and is my preferred specification, I find that a half-standard-deviation increment in business skills (about two additional years) is predicted to lower the probability of entering basic business ownership in the next year by 14.4% and increase the probability of entering advanced business ownership by 34.4%. These estimates—which, as seen in columns 1 and 3, are robust to the exclusion of specialized skills and/or balanced skill count—strongly support my argument that business skills are rewarded more in advanced business ownership than in wage employment, but relatively less in basic business ownership (prediction 1).

Clearly, the estimates just described also support my more general argument that it is important to distinguish between basic and advanced businesses. To underscore this argument, in panel B of table 7 I present estimated average marginal effects that are similar in all respects to those reported in panel A, except these estimates are based on a binomial logit model estimated for a pooled sample of basic and advanced businesses. Column 5 of panel B reveals that the same half-standard-deviation increment in business skills is predicted

to increase the probability of entering business ownership of either type by an imprecisely estimated 3.3%. Unsurprisingly, the strong positive and negative associations identified by the competing-risks MNL effectively offset each other.

Returning to the column 5 estimates in panel A of table 7, I find that a half-standard-deviation increase in specialized skills (about 2.5 years) is predicted to lower the probability of entering basic business ownership by 13.1% and lower the probability of entering advanced business ownership by a statistically insignificant 4.8%. When I use estimates from the binomial logit model (column 5 in panel B), the same increment is predicted to lower the probability of entering either type of business by 10.8%. These findings do not mean specialized skills are unimportant to business owners. Instead, they suggest that specialized skills yield an even higher return in wage employment than in either type of business ownership, thus increasing the opportunity cost of transitioning from wage employment into business ownership. In short, they support prediction 2.

My argument that skill balance does not predict entry into business ownership conditional on business skills and specialized skills is supported by the estimates in panel A, column 5 of table 7. A half-standard-deviation increment in the balanced skill count is predicted to raise the probabilities of entering basic and advanced business ownership by only 1.8% and 1.0%, respectively, and neither estimate is statistically significant. Skill balance does predict entry into advanced business ownership when business skills and specialized skills are excluded (column 4): the same increment is predicted to raise that probability by 7.6%, as the JOAT hypothesis predicts. Because workers with more balanced skills also tend to have accumulated more business skills (the partial correlation is 0.21 in the entry sample; see table 6), the balanced skill count partly stands in for business skills in column 4, where they are not controlled for. Together, these estimates show that skill balance does not predict entry into either type of business ownership once business skills and specialized skills are accounted for, in contrast to the JOAT hypothesis.

My finding that skill balance does not predict entry into business ownership stands in contrast to the findings of many prior studies, in which skill balance *does* predict entry, typically when all business owners are pooled and skill balance is measured by the number of occupations or employers a worker has held. To reconcile these findings, I replicate the conventional test in my data by pooling basic and advanced business ownership in a binomial logit model that includes occupation count but none of my skill measures (column 6 of panel B in table 7). This replication reproduces the evidence in support of the JOAT hypothesis: a half-standard-deviation increment in the number of occupations held to date is predicted to raise the probability of entering business ownership by 11.0%. However, the MNL estimates in column 6 of panel A reveal that this result comes almost entirely

from basic business ownership: the same increment is predicted to raise the probability of entering basic business ownership by 14.8% but the probability of entering advanced business ownership by a statistically insignificant 1.2%. Moreover, when I add my three skill measures (column 7), occupation count still predicts a 12.9% higher probability of entering basic business ownership. Because my direct measure of skill balance is now in the model and itself has no predictive power, this remaining association is unlikely to reflect skill balance. The same reasoning applies to employer count, which also rises with every job change, regardless of whether the new job uses different skills.

I argue that what occupation count captures instead is job churning. The number of occupations held to date rises with any change of occupation, including changes driven by poor performance or involuntary job loss. To test this argument, I add the number of involuntary job losses to the model in column 8 of panel A in table 7. If occupation count predicts entry because it reflects job churning, its estimated marginal effect should fall once involuntary job losses are controlled for. As expected, the estimated marginal effect of occupation count on entry into basic business ownership falls by approximately a third, from 12.9% in column 7 to 8.4% in column 8. Involuntary job losses themselves are also strong predictors of entry: a half-standard-deviation increment is predicted to raise the probability of entering basic business ownership by 11.3% and the probability of entering advanced business ownership by 6.5%. The larger estimate for basic business ownership is consistent with table 4, which shows that basic entrants have experienced almost twice as many involuntary job losses per year as advanced entrants. Together with the previous paragraph, these findings support my argument that existing evidence for the JOAT hypothesis based on occupation count is misleading: in my data, the association between occupation count and entry into business ownership comes almost entirely from basic business ownership, does not reflect skill balance, and is explained in part by job churning.[24]

As discussed in section 5.3, the most likely source of endogeneity in my entry model is that workers with a strong taste for business ownership accumulate more business skills in anticipation of owning a business, which would bias the estimated effect of business skills

[24]My findings also help resolve two puzzles in the JOAT literature. If the number of occupations a worker has held measured skill balance, the JOAT hypothesis would predict that it matters most for owners of businesses complex enough to require coordinating multiple functions, and that it raises the income of business owners. However, Lechmann and Schnabel (2014) find that skill balance, measured by the number of occupations a worker has held, predicts entry only into business ownership without employees, and Åstebro and Thompson (2011) find that skill balance, measured by the number of occupations and industries a worker has held, is associated with lower household income, especially among business owners. Both findings are consistent with mine if business ownership without employees more closely resembles basic business ownership: occupation count predicts entry mainly into basic business ownership, where earnings are substantially lower than in advanced business ownership (table 5), and it partly reflects job churning rather than skill balance.

upward. I cannot control for these tastes directly, but I can assess how much my key estimates change when I control for observed characteristics that are likely to be related to them. In table 8, I add groups of baseline regressors one at a time, starting with experience alone in column 1 and reaching my preferred specification in column 7; the two groups most closely related to tastes for business ownership enter last: personality traits (including risk tolerance) and whether a family member owned a business. Column 8 adds the most direct proxy for these tastes, whether the respondent considers herself an entrepreneur. I do not include this variable in my preferred specification because the question was first asked in 2010, after many respondents had already owned a business, so their answers might partly reflect that experience rather than a prior taste. If these observed proxies for tastes change the estimates little, unobserved tastes are also unlikely to change them much. The estimated marginal effect of business skills on the probability of entering advanced business ownership remains large and statistically significant in every column. Among the baseline regressors, education and AFQT percentile scores change it the most, from 49.2% in column 3 to 40.2% in column 4, whereas personality traits and family business background together lower it only from 37.3% in column 5 to 34.4% in column 7. Considering oneself an entrepreneur, the most direct proxy for these tastes, lowers it further to 23.6% in column 8. For basic business ownership, the estimate becomes more negative as these proxies for tastes are added, from -12.2% in column 5 to -14.4% in column 7 and -17.7% in column 8. The fact that these proxies move the estimated marginal effects of business skills for both types of business ownership in the same direction also indicates the direction of the bias: tastes for business ownership are positively correlated with business skills regardless of the type of business, and so bias both estimates upward. Such a bias cannot account for my finding that business skills promote entry into advanced business ownership but detract from entry into basic business ownership; if anything, my estimate understates the negative effect of business skills on entry into basic business ownership. Although this exercise cannot rule out bias from unobserved tastes, it shows that the positive association between business skills and entry into advanced business ownership and their negative association with entry into basic business ownership are both robust to controlling for observed proxies for these tastes.

6.2 Business Revenue and Earnings

Before turning to the estimates for my business revenue and earnings models, I again state the arguments they are designed to test. Whether a worker enters business ownership depends on how a skill is rewarded in business ownership relative to wage employment, but once she owns a business, her revenue and earnings depend on how the skill is rewarded within business ownership. As a result, a skill that lowers the probability of entry can still raise

revenue and earnings among those who enter. First, because I argue that business owners need business skills and specialized skills rather than a balance across all skills, I expect both business skills and specialized skills to raise revenue and earnings in both types of business ownership (prediction 3), even though specialized skills lower the probability of entering either type, and I expect my direct measure of skill balance to have little association with either outcome. Second, because business skills are especially valuable when running an advanced business, I expect them to raise revenue and earnings more in advanced than in basic business ownership (prediction 3); for this reason, I estimate the revenue and earnings models separately for the two types.

In table 9, I report estimated average marginal effects of my key skill measures on business revenue and earnings, based on the estimates in appendix table A5, separately for basic business ownership (panel A) and advanced business ownership (panel B). Column 1 reports Tobit estimates for log annual revenue, and columns 2 and 3 report OLS estimates for log weekly earnings, first for the same businesses and years as column 1 and then for the full earnings sample. Focusing first on revenue in column 1, I find that a half-standard-deviation increment in business skills is predicted to raise revenue by 25.2% for basic business owners and by 41.9% for advanced business owners. These estimates support my argument that all business owners benefit from business skills, but that the payoff is larger in advanced business ownership (prediction 3). The estimates for weekly earnings are smaller and less precise, but in the full earnings sample (column 3) they follow the same pattern: business skills are predicted to raise earnings by 13.5% for advanced business owners, an estimate that is significant at the 10% level, compared with a statistically insignificant 4.1% for basic business owners.

Turning to specialized skills, I find that a half-standard-deviation increment is predicted to raise revenue by 16.2% for basic business owners (column 1 of panel A in table 9) and their weekly earnings by 14.0% and 18.3% (columns 2 and 3). Together with the entry estimates in section 6.1, these estimates show that specialized skills lower the probability of entering basic business ownership, likely because they are even more valuable in wage employment, but substantially raise the revenue and earnings of those who do enter. For advanced business owners (panel B), the estimates are also positive but smaller: the same increment is predicted to raise revenue by 6.7% and earnings by 6.1% in both earnings samples, although none of these estimates is statistically significant. These positive estimates for both types of business ownership support prediction 3. The larger estimates for basic business owners than for advanced business owners are consistent with my distinction between the two types of business ownership: owners of basic businesses rely primarily on their specialized skills, whereas owners of advanced businesses also rely heavily on business skills because they must

manage multiple functions, such as financing, hiring, and marketing. Unlike the estimates for business skills, which are much larger for revenue than for earnings, the estimates for specialized skills are similar for the two outcomes. This difference suggests that business skills pay off mainly by growing the business, possibly through reinvested profits, whereas an owner's expertise in her line of work pays off directly in the earnings she takes home.

My direct measure of skill balance has little association with business revenue or earnings conditional on business skills and specialized skills. In column 1 of panel A in table 9, a half-standard-deviation increment in the balanced skill count is predicted to raise the revenue of basic business owners by 6.8%, an estimate that is significant only at the 10% level, compared with 16.2% for the same increment in specialized skills. The estimates for their earnings are similarly small, at 4.1% and 3.3% in columns 2 and 3. For advanced business owners (panel B), none of the estimates for the balanced skill count is statistically significant. These findings show that skill balance, when measured directly, is not an important determinant of business owners' revenue and earnings, in contrast to the JOAT hypothesis.

7 Conclusion

In this study, I extend the JOAT literature in three ways. First, I argue that what prepares a worker for business ownership is not balance across all skills, but rather business skills and specialized skills relevant to the business. Second, because the skills that matter for business ownership are likely to depend on the type of business, I distinguish between two types of business ownership: basic and advanced. Advanced businesses are incorporated and/or start with higher levels of capital; all other businesses are basic. Third, I revisit conventional tests of the JOAT hypothesis to assess whether occupation count (one of the most common proxies for skill balance) predicts entry into business ownership because it captures skill balance or because it reflects job churning. I do so by replicating these tests with occupation count and examining how its estimated effect on entry into business ownership changes when my skill measures and the number of involuntary job losses are controlled for. To pursue these innovations, I develop a simple, static choice model of how business skills and specialized skills affect both the choice to enter each type of business ownership and the payoffs to these skills among business owners. I then combine nearly 40 years of NLSY79 work histories with O*NET data on the skill requirements of each occupation to estimate a competing-risks model of entry into basic and advanced business ownership, as well as models of business revenue and earnings among business owners.

My empirical findings show that each of these three innovations of the JOAT literature is warranted. First, what matters for business ownership is business skills and specialized skills, not balance across all skills. Once business skills and specialized skills are accounted for, my

direct measure of skill balance does not predict entry into either type of business ownership and has little association with revenue and earnings. In contrast, a half-standard-deviation increment in business skills is predicted to raise the probability of entering advanced business ownership by 34%, and both business skills and specialized skills are positively associated with the revenue and earnings of business owners. Specialized skills lower the probability of entering business ownership but raise revenue and earnings among those who enter, as my framework predicts: entry depends on how a skill is rewarded in business ownership relative to wage employment, whereas revenue and earnings depend on how it is rewarded within business ownership. Second, the estimated effects of business skills and specialized skills differ between basic and advanced business ownership. Business skills raise the probability of entering advanced business ownership but lower the probability of entering basic business ownership, a pattern that disappears when the two types are pooled. Among business owners, business skills are more strongly associated with revenue in advanced than in basic business ownership, whereas specialized skills are more strongly associated with revenue and earnings in basic than in advanced business ownership. Third, occupation count reflects job churning more than skill balance. When I replicate conventional tests, occupation count predicts entry into business ownership almost entirely through basic business ownership, and involuntary job losses explain a substantial part of this association. Together, these findings suggest that conventional tests of the JOAT hypothesis attribute to skill balance what in fact reflects business skills and job churning, and that they hide important differences between types of business ownership.

These findings have implications both for workers who hope to own a business and for programs that support them, and the implications depend on the type of business. For workers who hope to run an advanced business, experience in occupations that involve hiring and supervising workers, managing budgets, and marketing is likely to be more valuable than experience across many unrelated jobs, because business skills, rather than skill balance, predict both entry into advanced business ownership and higher revenue among advanced business owners. Several programs supported by the SBA already focus on building business skills: SBDCs help workers build business skills before they start a business, and the SBA's T.H.R.I.V.E. program offers executive-level training to owners of established, growing businesses. Owners of basic businesses also benefit from business skills, but specialized skills matter more for them than for owners of advanced businesses, so these owners might benefit from assistance in strengthening their expertise in their line of work, for example through mentoring from experienced people in the same field, as SCORE offers. Because involuntary job losses predict entry into basic business ownership, programs that serve displaced workers might be a natural place to offer this assistance. More broadly, all business owners benefit

from being jacks of business trades rather than jacks of all trades, as well as from specialized skills, but the payoffs to these skills differ by type of business.

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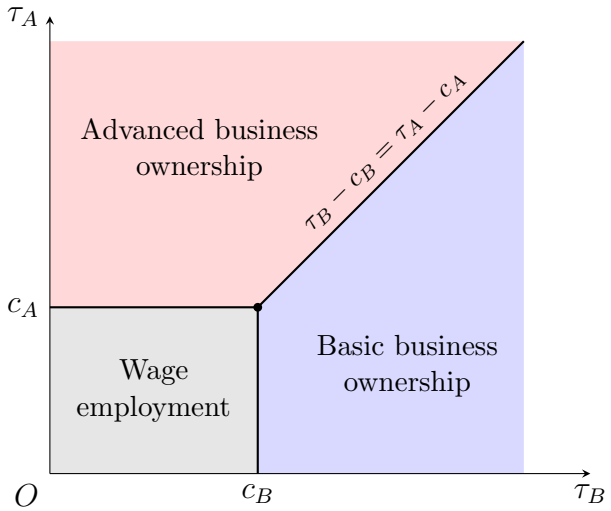
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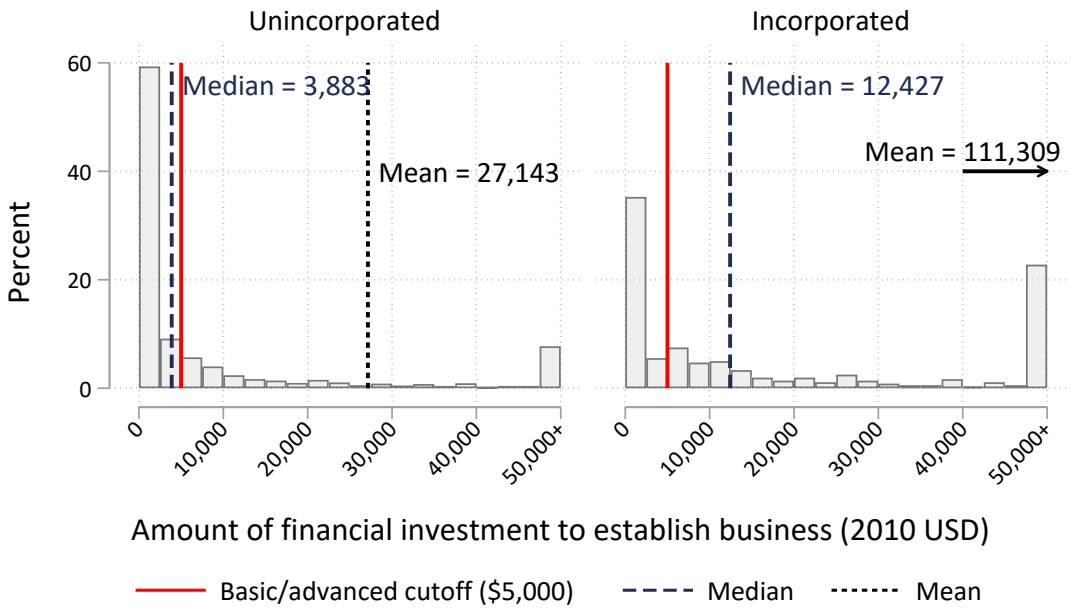
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Figure 1: The Worker's Choice Between Employment Options as a Function of Tastes for Business Ownership



Note: As defined in section 3.1, τ_B and τ_A are the worker's tastes for basic and advanced business ownership, respectively, while c_B and c_A are the opportunity costs of each business option. Each shaded region identifies the taste (τ_B, τ_A) combinations that lead to the given option being chosen, given the opportunity costs c_B and c_A.

Figure 2: Distribution of Start-Up Capital by Incorporation Status



Note: This figure shows the distribution of start-up capital at the start of each business, separately for incorporated and unincorporated businesses. The thick, solid (red) vertical line indicates the $5,000 cutoff (equal to the median across all businesses with positive start-up capital) that I use to distinguish basic from advanced businesses.

Table 1: Means and Standard Deviations for All Regressors in the Five Estimation Samples

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
Panel A: Key Independent Variables					
O*NET Business skills (years)	4.81	7.56	8.69	4.32	6.73
	(4.46)	(5.47)	(5.96)	(4.40)	(5.46)
O*NET Specialized skills (years)	4.92	6.17	7.06	3.74	5.49
	(5.34)	(5.52)	(5.74)	(4.30)	(5.13)
O*NET Balanced skill count	28.55	28.77	28.96	28.60	28.88
	(0.81)	(0.65)	(0.57)	(0.77)	(0.65)
Number of occupations held to date	4.58	–	–	–	–
	(2.44)				
Number of involuntary job losses to date	2.01	–	–	–	–
	(2.53)				
Panel B: Baseline Regressors—Work Experience					
Potential experience (years)	19.18	24.18	20.82	15.40	16.76
	(11.71)	(9.23)	(8.50)	(9.58)	(8.96)
Actual experience (years)	14.23	18.52	18.07	11.15	14.46
	(10.98)	(9.41)	(8.73)	(8.71)	(8.76)
Number of past business ownership spells	0.29	2.53	2.13	1.56	1.76
	(0.69)	(3.00)	(2.77)	(2.24)	(2.41)
Tenure in the business (years)	–	5.40	10.22	3.05	7.40
		(7.87)	(10.54)	(5.50)	(8.98)
Panel C: Baseline Regressors—Individual Characteristics					
1 if female[a]	0.53	0.47	0.25	0.46	0.27
Race/ethnicity[a]					
1 if Hispanic	0.19	0.16	0.16	0.16	0.16
1 if Black	0.32	0.26	0.18	0.24	0.16
1 if neither Black nor Hispanic	0.49	0.58	0.66	0.61	0.68
Marital status					
1 if never married	0.29	0.16	0.09	0.22	0.12
1 if currently married	0.56	0.53	0.70	0.56	0.70
1 if previously married	0.15	0.31	0.21	0.22	0.18
1 if cohabiting with an unmarried partner	0.10	0.04	0.04	0.06	0.04
1 if youngest child is aged six or younger	0.24	0.05	0.07	0.23	0.18
Education					
1 if less than high school	0.13	0.07	0.05	0.12	0.06
1 if high school graduate	0.45	0.39	0.36	0.43	0.39
1 if some college	0.24	0.22	0.23	0.22	0.24
1 if college or above	0.19	0.33	0.36	0.23	0.31
AFQT percentile score[a]	40.92	51.18	52.87	46.51	52.10

Continued on next page

Table 1 (continued)

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
	(28.53)	(29.87)	(28.04)	(29.44)	(27.63)
Risk tolerance					
1 if least risk tolerant	0.52	0.45	0.41	0.46	0.46
1 if less risk tolerant	0.14	0.20	0.22	0.15	0.16
1 if more risk tolerant	0.14	0.12	0.11	0.14	0.13
1 if most risk tolerant	0.20	0.22	0.26	0.24	0.25
1 if unknown	0.00	0.01	0.00	0.01	0.00
Sociability at age six[a]					
1 if extremely shy	0.18	0.15	0.12	0.14	0.12
1 if somewhat shy	0.47	0.49	0.43	0.52	0.45
1 if somewhat outgoing	0.24	0.24	0.31	0.23	0.28
1 if extremely outgoing	0.11	0.12	0.15	0.12	0.15
Rotter locus of control score (4–16)	7.85	7.13	6.86	7.34	7.01
	(2.37)	(2.12)	(2.16)	(2.13)	(2.14)
Extraverted, enthusiastic (1–7)[a]	5.03	5.18	5.30	5.15	5.28
	(1.68)	(1.51)	(1.51)	(1.44)	(1.49)
Open to new experiences, complex (1–7)[a]	5.23	5.32	5.39	5.31	5.35
	(1.57)	(1.44)	(1.44)	(1.34)	(1.41)
Conventional, uncreative (1–7)[a]	3.32	2.97	3.04	3.02	3.03
	(1.85)	(1.67)	(1.73)	(1.63)	(1.71)
1 if considers oneself an entrepreneur[a,b]	0.15	–	–	–	–
Panel D: Baseline Regressors—Family Background					
Net family wealth (thousands of $)	158.29	–	–	–	–
	(394.36)				
Net family income (thousands of $)	64.20	–	–	–	–
	(79.59)				
1 if father had at least some college[a]	0.19	0.29	0.30	0.25	0.30
1 if father was a professional or manager in 1979[a]	0.16	0.25	0.30	0.22	0.30
1 if any family member owned a business[a]	0.26	0.35	0.43	0.34	0.44
Panel E: Region and Industry					
Region					
1 if Northeast	–	0.14	0.14	0.15	0.14
1 if North Central	–	0.25	0.26	0.25	0.27
1 if South	–	0.39	0.37	0.37	0.36
1 if West	–	0.20	0.21	0.22	0.21
1 if unknown	–	0.01	0.02	0.01	0.02
Industry					

Continued on next page

Table 1 (continued)

	Business Ownership	Business Revenue		Weekly Earnings	
	Entry	Basic	Advanced	Basic	Advanced
	(1)	(2)	(3)	(4)	(5)
1 if mining and construction	–	0.16	0.17	0.16	0.17
1 if manufacturing	–	0.03	0.07	0.03	0.08
1 if transportation, communications, and public utilities	–	0.04	0.08	0.03	0.08
1 if wholesale and retail trade	–	0.08	0.15	0.09	0.17
1 if services	–	0.32	0.26	0.40	0.27
1 if professional services	–	0.28	0.17	0.19	0.15
1 if other	–	0.09	0.08	0.09	0.08
Number of person-year observations	251,214	2,430	2,465	10,383	5,938
Number of individuals	7,116	938	832	2,370	1,103

Note: Summary statistics are computed over the person-years of each estimation sample. In the business revenue and weekly earnings samples, the skill measures and experience are measured at the start of each business. See section 4 for details. A dash means that the regressor is not included in the model for that outcome.

[a] Time-invariant; all other variables are time-varying.

[b] I use this variable only in column 8 of table 8.

Table 2: Distributions of Number and Type of Business Ownership

	N. individuals	Share (%)
Panel A: Distribution of Number of Businesses Owned per Individual		
0	4,166	58.5
1	1,838	25.8
2	761	10.7
3	240	3.4
4	81	1.1
5	20	0.3
6	7	0.1
7	1	0.0
8	2	0.0
Total	7,116	100.0
Panel B: Distribution of Overall Business Ownership Pattern		
Never entered	4,166	58.5
Basic only	1,989	28.0
Advanced only	596	8.4
Both	365	5.1
Total	7,116	100.0

Table 3: The Eight Most Common Occupations by Type of Business

	Basic Business	Advanced Business
1	Childcare Workers	Managers, All Other
2	Landscaping and Groundskeeping Workers	Sales Workers
3	Sales Workers	Driver/Sales Workers and Truck Drivers
4	Managers, All Other	First-Line Supervisors of Retail Sales Workers
5	Maids and Housekeeping Cleaners	Carpenters
6	Carpenters	Bookkeeping, Accounting, and Auditing Clerks
7	Painters, Construction and Maintenance	Chief Executives
8	Construction Laborers	Food Service Managers

Note: The eight most common occupations are listed in order of frequency, based on 3,443 basic businesses and 1,108 advanced businesses owned by individuals in the entry sample. Each business is assigned the owner's 6-digit SOC occupation at entry.

Table 4: Means and Standard Deviations of Select Individual Characteristics at Time of Entry into Basic and Advanced Businesses

	Basic (1)	Advanced (2)
Age (years)	34.99	38.50
	(10.77)	(9.84)
Potential experience (years)	15.46	18.09
	(10.68)	(10.07)
Actual experience (years)	10.86	15.33
	(9.61)	(9.81)
O*NET Business skills (years)	3.60	5.96
	(3.87)	(4.69)
O*NET Business skills per year of actual experience[a]	0.31	0.38
	(0.11)	(0.13)
O*NET Specialized skills (years)	3.24	5.10
	(3.92)	(4.52)
O*NET Specialized skills per year of actual experience[a]	0.30	0.33
	(0.18)	(0.19)
O*NET Balanced skill count	28.56	28.83
	(0.78)	(0.69)
Number of occupations held to date	4.77	4.96
	(2.56)	(2.44)
Number of occupations held per year of actual experience[a]	0.71	0.48
	(0.60)	(0.47)
Number of involuntary job losses per year of actual experience[a]	0.32	0.17
	(0.45)	(0.26)
Number of past business ownership spells per year of actual experience[a]	0.07	0.05
	(0.14)	(0.09)
Education		
1 if less than high school	0.16	0.08
1 if high school graduate	0.44	0.37
1 if some college	0.22	0.24
1 if college or above	0.18	0.31
AFQT percentile score	43.05	50.99
	(28.93)	(28.22)
Number of person-year observations	3,463	1,106

Note: Summary statistics are calculated using the last person-year observations before entry into basic and advanced business ownership.

[a] Per-year measures exclude person-years with less than one year of actual experience.

Table 5: Means and Standard Deviations of Select Business Characteristics by Business Type

	Basic (1)	Advanced (2)
Annual revenue, excluding top-coded values (thousands of $)[a]	63.32	215.48
	(291.10)	(439.64)
1 if revenue is top-coded[a]	0.01	0.09
Weekly earnings ($)	443.47	843.63
	(717.63)	(1,198.74)
Hours worked per week	31.91	40.86
	(19.82)	(20.95)
Business duration (years)	3.93	7.61
	(5.50)	(7.74)
1 if business lasted less than 1 year	0.30	0.13
1 if business lasted 5 years or more	0.23	0.49
1 if business is ongoing at the last interview	0.12	0.32
Occupation group		
1 if management, business, science, and arts	0.25	0.42
1 if service	0.34	0.11
1 if sales and office	0.14	0.21
1 if natural resources, construction, and maintenance	0.19	0.15
1 if production, transportation, and material moving	0.08	0.11
Number of businesses, revenue sample	1,221	1,027
Number of businesses, earnings sample	4,118	1,554

[a] These statistics are from the revenue sample; all other rows are from the earnings sample.

Table 6: Partial Correlations Among Skill and Count Measures

	(1)	(2)	(3)	(4)	(5)
Panel A: Entry Sample, Person-Years					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.396***	1.000			
(3) O*NET Balanced skill count	0.214***	-0.086***	1.000		
(4) Number of occupations held to date	-0.105***	-0.222***	0.249***	1.000	
(5) Number of involuntary job losses to date	-0.241***	-0.167***	0.091***	0.476***	1.000
Number of person-year observations = 251,214					
Panel B: Basic Businesses in the Earnings Sample, at Business Start					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.451***	1.000			
(3) O*NET Balanced skill count	0.145***	-0.026*	1.000		
(4) Number of occupations held to date	-0.134***	-0.194***	0.252***	1.000	
(5) Number of involuntary job losses to date	-0.129***	-0.125***	0.128***	0.358***	1.000
Number of businesses = 4,118					
Panel C: Advanced Businesses in the Earnings Sample, at Business Start					
(1) O*NET Business skills (years)	1.000				
(2) O*NET Specialized skills (years)	0.559***	1.000			
(3) O*NET Balanced skill count	0.158***	0.012	1.000		
(4) Number of occupations held to date	-0.193***	-0.262***	0.182***	1.000	
(5) Number of involuntary job losses to date	-0.232***	-0.208***	0.067***	0.392***	1.000
Number of businesses = 1,554					

Note: Correlations control for actual experience within each panel. *p < .10; **p < .05; ***p < .01.

Table 7: Estimated Average Marginal Effects of Key Regressors on the Probability of Business Ownership Entry (Based on MNL and Logit Estimates Shown in Table A3)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.192***		-0.135***		-0.144***		-0.122***	-0.069
	(0.040)		(0.041)		(0.043)		(0.044)	(0.045)
Std. O*NET Specialized skills		-0.152***	-0.136***		-0.131***		-0.108***	-0.114***
		(0.021)	(0.022)		(0.022)		(0.023)	(0.023)
Std. O*NET Balanced skill count				0.013	0.018		-0.010	-0.012
				(0.014)	(0.015)		(0.015)	(0.015)
Std. Number of occupations held to date						0.148***	0.129***	0.084***
						(0.013)	(0.015)	(0.017)
Std. Number of involuntary job losses to date								0.113***
								(0.020)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.345***		0.350***		0.344***		0.340***	0.362***
	(0.063)		(0.066)		(0.068)		(0.069)	(0.070)
Std. O*NET Specialized skills		0.010	-0.049		-0.048		-0.050	-0.052
		(0.032)	(0.032)		(0.032)		(0.032)	(0.032)
Std. O*NET Balanced skill count				0.076***	0.010		0.014	0.013
				(0.024)	(0.026)		(0.026)	(0.026)
Std. Number of occupations held to date						0.012	-0.015	-0.037
						(0.023)	(0.026)	(0.028)
Std. Number of involuntary job losses to date								0.065**
								(0.030)

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Table 7 (continued)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Binary Logit								
Any Business Ownership (Unconditional Prob. = 0.018)								
Std. O*NET Business skills	-0.006		0.036		0.033		0.052	0.098***
	(0.034)		(0.035)		(0.036)		(0.037)	(0.038)
Std. O*NET Specialized skills		-0.101***	-0.112***		-0.108***		-0.093***	-0.098***
		(0.018)	(0.019)		(0.019)		(0.019)	(0.019)
Std. O*NET Balanced skill count				0.028**	0.011		-0.009	-0.011
				(0.012)	(0.013)		(0.013)	(0.013)
Std. Number of occupations held to date						0.110***	0.091***	0.052***
						(0.012)	(0.013)	(0.015)
Std. Number of involuntary job losses to date								0.101***
								(0.017)
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: In both panels, the reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each skill and count measure. See table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table 8: Sensitivity to Additional Controls: Estimated Average Marginal Effects of Key Regressors on the Probability of Business Ownership Entry (Based on MNL Estimates Shown in Table A4)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.066	-0.085**	-0.087**	-0.110***	-0.122***	-0.139***	-0.144***	-0.177***
	(0.041)	(0.040)	(0.040)	(0.043)	(0.043)	(0.043)	(0.043)	(0.043)
Std. O*NET Specialized skills	-0.127***	-0.138***	-0.132***	-0.134***	-0.133***	-0.132***	-0.131***	-0.118***
	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)	(0.022)
Std. O*NET Balanced skill count	0.052***	0.041***	0.023	0.020	0.021	0.020	0.018	0.008
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.014)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.424***	0.426***	0.492***	0.402***	0.373***	0.351***	0.344***	0.236***
	(0.064)	(0.065)	(0.065)	(0.068)	(0.068)	(0.068)	(0.068)	(0.066)
Std. O*NET Specialized skills	-0.074**	-0.086***	-0.043	-0.047	-0.045	-0.050	-0.048	-0.028
	(0.031)	(0.033)	(0.033)	(0.032)	(0.032)	(0.032)	(0.032)	(0.034)
Std. O*NET Balanced skill count	0.123***	0.108***	0.015	0.013	0.013	0.014	0.010	0.003
	(0.026)	(0.026)	(0.027)	(0.026)	(0.026)	(0.026)	(0.026)	(0.025)
Potential and actual experience	✓	✓	✓	✓	✓	✓	✓	✓
Past business ownership spells		✓	✓	✓	✓	✓	✓	✓
Gender, race, and household composition			✓	✓	✓	✓	✓	✓
Education and AFQT				✓	✓	✓	✓	✓
Wealth, income, and father's background					✓	✓	✓	✓
Risk tolerance and personality						✓	✓	✓
Family member owned a business							✓	✓
Considers oneself an entrepreneur								✓
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each skill measure. Columns 2–8 progressively add the control groups indicated at the bottom of the table. See table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table 9: Estimated Average Marginal Effects of Business Skills, Specialized Skills, and Skill Balance on Business Revenue and Weekly Earnings in Business Ownership (Based on Estimates Shown in Table A5)

	Revenue Sample		Earnings Sample
	Log Annual Revenue (Tobit)	Log Weekly Earnings (OLS)	Log Weekly Earnings (OLS)
	(1)	(2)	(3)
Panel A: Basic Business Ownership			
Std. O*NET Business skills	0.252**	0.138	0.041
	(0.108)	(0.106)	(0.081)
Std. O*NET Specialized skills	0.162***	0.140***	0.183***
	(0.048)	(0.054)	(0.043)
Std. O*NET Balanced skill count	0.068*	0.041	0.033**
	(0.042)	(0.043)	(0.016)
Tenure in the business	0.153***	0.131***	0.097***
	(0.014)	(0.014)	(0.008)
Number of person-year observations	2,430	2,430	10,383
Number of businesses	1,221	1,221	4,118
Mean business revenue or earnings	56,859	412	514
Panel B: Advanced Business Ownership			
Std. O*NET Business skills	0.419***	0.094	0.135*
	(0.115)	(0.093)	(0.078)
Std. O*NET Specialized skills	0.067	0.061	0.061
	(0.068)	(0.061)	(0.049)
Std. O*NET Balanced skill count	0.013	0.044	0.036
	(0.062)	(0.046)	(0.029)
Tenure in the business	0.106***	0.100***	0.089***
	(0.011)	(0.010)	(0.007)
Number of person-year observations	2,465	2,465	5,938
Number of businesses	1,027	1,027	1,554
Mean business revenue or earnings	215,578	1,088	1,088

Note: Estimated marginal effects correspond to a half-standard-deviation increase in each skill and are in log points. Mean revenue in column 1 excludes top-coded reports. See table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Appendix A: Additional Tables

Table A1: Classification of O*NET Skill and Knowledge Elements by Skill Type

Skill Type	Skill Elements	Knowledge Elements
Core Business	Service Orientation	Economics and Accounting
		Sales and Marketing
		Customer and Personal Service
		Production and Processing
Managerial	Management of Financial Resources	Administration and Management
	Management of Material Resources	Personnel and Human Resources
	Management of Personnel Resources	
Interpersonal	Social Perceptiveness	
	Coordination	
	Persuasion	
	Negotiation	
	Instructing	
Other	Reading Comprehension	Clerical
	Active Listening	Transportation
	Writing	Food Production
	Speaking	Computers and Electronics
	Mathematics	Engineering and Technology
	Science	Design
	Critical Thinking	Building and Construction
	Active Learning	Mechanical
	Learning Strategies	Mathematics
	Monitoring	Physics
	Complex Problem Solving	Chemistry
	Operations Analysis	Biology
	Technology Design	Psychology
	Equipment Selection	Sociology and Anthropology
	Installation	Geography
	Programming	Medicine and Dentistry
	Operation Monitoring	Therapy and Counseling
	Operation and Control	Education and Training
	Equipment Maintenance	English Language
	Troubleshooting	Foreign Language
	Repairing	Fine Arts
	Quality Control Analysis	History and Archeology
	Judgment and Decision Making	Philosophy and Theology
	Systems Analysis	Public Safety and Security
	Systems Evaluation	Law and Government
	Time Management	Telecommunications
		Communications and Media

Note: This table shows my classification of all 68 O*NET skill and knowledge elements as core business, managerial, interpersonal, or other. I use the 15 elements not classified as "other" to construct my business skills measure.

Table A2: Estimated Average Marginal Effects of General Skill Requirements and Key Regressors on the Probability of Business Ownership Entry

	Dependent Variable: Entry into Business Ownership		
	(1)	(2)	(3)
Multinomial Logit			
Basic Business Ownership (Unconditional Prob. = 0.014)			
Std. O*NET Business skills	-0.144***		-0.128***
	(0.043)		(0.047)
Std. O*NET Specialized skills	-0.131***		-0.127***
	(0.022)		(0.022)
Std. O*NET Balanced skill count	0.018		0.019
	(0.015)		(0.015)
Std. Actual experience in Job Zone 4+ occupations		-0.071***	-0.030
		(0.025)	(0.029)
Advanced Business Ownership (Unconditional Prob. = 0.004)			
Std. O*NET Business skills	0.344***		0.353***
	(0.068)		(0.073)
Std. O*NET Specialized skills	-0.048		-0.047
	(0.032)		(0.033)
Std. O*NET Balanced skill count	0.010		0.012
	(0.026)		(0.026)
Std. Actual experience in Job Zone 4+ occupations		0.090***	-0.019
		(0.035)	(0.040)
Number of individuals	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214

Note: This table assesses whether my measure of business skills, BX_{it}, reflects general skill requirements. Column 1 repeats column 5 of table 7. Job Zone 4+ occupations are those in O*NET Job Zones 4 and 5, which require the most preparation. The reference category is staying out of business ownership. Estimated marginal effects are the average percentage change in the predicted probability of entry from a half-standard-deviation increase in each measure. See table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A3: MNL and Logit Coefficient Estimates for Entry into Business Ownership

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.195***		-0.138***		-0.146***		-0.124***	-0.069
	(0.041)		(0.042)		(0.044)		(0.045)	(0.046)
Std. O*NET Business skills^2	0.022***		0.028***		0.028***		0.025***	0.021***
	(0.005)		(0.006)		(0.006)		(0.006)	(0.006)
Std. O*NET Specialized skills		-0.154***	-0.138***		-0.132***		-0.109***	-0.115***
		(0.022)	(0.022)		(0.023)		(0.024)	(0.024)
Std. O*NET Specialized skills^2		-0.004	-0.012*		-0.012		-0.012	-0.012
		(0.006)	(0.008)		(0.007)		(0.007)	(0.007)
Std. O*NET Balanced skill count				0.014	0.018		-0.010	-0.012
				(0.014)	(0.015)		(0.015)	(0.015)
Std. O*NET Balanced skill count^2				-0.005*	-0.004		-0.004	-0.004
				(0.003)	(0.003)		(0.003)	(0.003)
Std. Number of occupations held to date						0.151***	0.132***	0.086***
						(0.014)	(0.016)	(0.018)
Std. Number of occupations held to date^2						-0.011***	-0.010**	-0.009**
						(0.004)	(0.004)	(0.004)
Std. Number of involuntary job losses to date								0.115***
								(0.020)
Std. Number of involuntary job losses to date^2								-0.008**
								(0.004)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.345***		0.350***		0.344***		0.340***	0.363***
	(0.064)		(0.066)		(0.069)		(0.069)	(0.071)
Std. O*NET Business skills^2	-0.031***		-0.013		-0.013		-0.013	-0.015
	(0.010)		(0.011)		(0.011)		(0.011)	(0.011)
Std. O*NET Specialized skills		0.009	-0.049		-0.048		-0.050	-0.052
		(0.032)	(0.033)		(0.033)		(0.033)	(0.033)
Std. O*NET Specialized skills^2		-0.042***	-0.048***		-0.047***		-0.048***	-0.048***
		(0.010)	(0.011)		(0.011)		(0.011)	(0.012)
Std. O*NET Balanced skill count				0.077***	0.010		0.014	0.013
				(0.024)	(0.026)		(0.026)	(0.026)
Std. O*NET Balanced skill count^2				0.007	-0.002		-0.001	-0.002
				(0.005)	(0.007)		(0.007)	(0.007)
Std. Number of occupations held to date						0.014	-0.013	-0.036
						(0.023)	(0.026)	(0.028)
Std. Number of occupations held to date^2						-0.003	-0.001	-0.000
						(0.006)	(0.007)	(0.007)
Std. Number of involuntary job losses to date								0.067**
								(0.030)
Std. Number of involuntary job losses to date^2								-0.006
								(0.004)

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Table A3 (continued)

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel B: Binary Logit								
Any Business Ownership (Unconditional Prob. = 0.018)								
Std. O*NET Business skills	-0.007		0.036		0.033		0.053	0.099**
	(0.034)		(0.035)		(0.037)		(0.038)	(0.039)
Std. O*NET Business skills^2	0.005		0.015***		0.015***		0.013**	0.009
	(0.005)		(0.005)		(0.005)		(0.006)	(0.006)
Std. O*NET Specialized skills		-0.102***	-0.113***		-0.109***		-0.094***	-0.099***
		(0.019)	(0.019)		(0.019)		(0.020)	(0.020)
Std. O*NET Specialized skills^2		-0.014***	-0.023***		-0.022***		-0.022***	-0.022***
		(0.005)	(0.006)		(0.006)		(0.006)	(0.006)
Std. O*NET Balanced skill count				0.028**	0.011		-0.009	-0.011
				(0.012)	(0.013)		(0.013)	(0.013)
Std. O*NET Balanced skill count^2				-0.002	-0.004		-0.004	-0.004
				(0.003)	(0.003)		(0.003)	(0.003)
Std. Number of occupations held to date						0.112***	0.093***	0.053***
						(0.012)	(0.014)	(0.015)
Std. Number of occupations held to date^2						-0.009***	-0.008**	-0.008**
						(0.003)	(0.004)	(0.004)
Std. Number of involuntary job losses to date								0.103***
								(0.018)
Std. Number of involuntary job losses to date^2								-0.008**
								(0.003)
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The models are those in table 7, with the same controls. All reported measures are standardized so that a one-unit increase equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A4: Sensitivity to Additional Controls: MNL Coefficient Estimates for Entry into Business Ownership

	Dependent Variable: Entry into Business Ownership							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Multinomial Logit								
Basic Business Ownership (Unconditional Prob. = 0.014)								
Std. O*NET Business skills	-0.067	-0.086**	-0.087**	-0.111**	-0.124***	-0.141***	-0.146***	-0.181***
	(0.042)	(0.040)	(0.041)	(0.043)	(0.044)	(0.044)	(0.044)	(0.043)
Std. O*NET Business skills^2	0.025***	0.023***	0.023***	0.025***	0.026***	0.027***	0.028***	0.028***
	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
Std. O*NET Specialized skills	-0.128***	-0.140***	-0.134***	-0.136***	-0.135***	-0.134***	-0.132***	-0.120***
	(0.022)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)	(0.023)
Std. O*NET Specialized skills^2	-0.019**	-0.009	-0.011	-0.011	-0.012	-0.012	-0.012	-0.013*
	(0.008)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Std. O*NET Balanced skill count	0.053***	0.042***	0.024	0.020	0.021	0.020	0.018	0.009
	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)	(0.015)
Std. O*NET Balanced skill count^2	-0.001	-0.002	-0.003	-0.004	-0.004	-0.004	-0.004	-0.005
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Advanced Business Ownership (Unconditional Prob. = 0.004)								
Std. O*NET Business skills	0.425***	0.427***	0.494***	0.403***	0.373***	0.351***	0.344***	0.234***
	(0.065)	(0.065)	(0.065)	(0.068)	(0.069)	(0.069)	(0.069)	(0.067)
Std. O*NET Business skills^2	-0.014	-0.016	-0.024**	-0.016	-0.017	-0.015	-0.013	-0.008
	(0.010)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Std. O*NET Specialized skills	-0.074**	-0.087***	-0.044	-0.047	-0.045	-0.050	-0.048	-0.028
	(0.032)	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)	(0.033)	(0.035)
Std. O*NET Specialized skills^2	-0.045***	-0.036***	-0.045***	-0.046***	-0.048***	-0.046***	-0.047***	-0.046***
	(0.011)	(0.011)	(0.012)	(0.011)	(0.011)	(0.011)	(0.011)	(0.011)
Std. O*NET Balanced skill count	0.124***	0.109***	0.015	0.013	0.013	0.015	0.010	0.003
	(0.026)	(0.026)	(0.027)	(0.026)	(0.026)	(0.026)	(0.026)	(0.026)
Std. O*NET Balanced skill count^2	0.009	0.008	-0.002	-0.002	-0.002	-0.001	-0.002	-0.001
	(0.005)	(0.005)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Potential and actual experience	✓	✓	✓	✓	✓	✓	✓	✓
Past business ownership spells		✓	✓	✓	✓	✓	✓	✓
Gender, race, and household composition			✓	✓	✓	✓	✓	✓
Education and AFQT				✓	✓	✓	✓	✓
Wealth, income, and father's background					✓	✓	✓	✓
Risk tolerance and personality						✓	✓	✓
Family member owned a business							✓	✓
Considers oneself an entrepreneur								✓
Number of individuals	7,116	7,116	7,116	7,116	7,116	7,116	7,116	7,116
Number of person-year observations	251,214	251,214	251,214	251,214	251,214	251,214	251,214	251,214

Note: The models are those in table 8, with the same controls. All reported measures are standardized so that a one-unit change equals half a standard deviation. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table A5: Tobit and OLS Coefficient Estimates for Business Revenue and Weekly Earnings in Business Ownership

	Revenue Sample		Earnings Sample
	Log Annual Revenue (Tobit)	Log Weekly Earnings (OLS)	Log Weekly Earnings (OLS)
	(1)	(2)	(3)
Panel A: Basic Business Ownership			
Std. O*NET Business skills	0.240**	0.125	0.027
	(0.103)	(0.102)	(0.063)
Std. O*NET Business skills^2	-0.030**	-0.031**	-0.005
	(0.015)	(0.014)	(0.011)
Std. O*NET Specialized skills	0.160***	0.140***	0.174***
	(0.046)	(0.052)	(0.035)
Std. O*NET Specialized skills^2	-0.007	0.003	-0.004
	(0.011)	(0.011)	(0.011)
Std. O*NET Balanced skill count	0.076*	0.045	0.036**
	(0.044)	(0.045)	(0.018)
Std. O*NET Balanced skill count^2	0.013**	0.006	0.002
	(0.006)	(0.006)	(0.002)
Tenure in the business	0.194***	0.167***	0.113***
	(0.020)	(0.021)	(0.010)
Tenure in the business^2	-0.004***	-0.003***	-0.003***
	(0.001)	(0.001)	(0.000)
Number of person-year observations	2,430	2,430	10,383
Number of businesses	1,221	1,221	4,118
Adjusted R^2 (pseudo R^2 for Tobit)	0.061	0.179	0.187
Mean business revenue or earnings	56,859	412	514
Panel B: Advanced Business Ownership			
Std. O*NET Business skills	0.439***	0.107	0.104
	(0.120)	(0.097)	(0.072)
Std. O*NET Business skills^2	-0.050***	-0.032**	-0.031***
	(0.017)	(0.015)	(0.012)
Std. O*NET Specialized skills	0.065	0.052	0.078*
	(0.071)	(0.063)	(0.044)
Std. O*NET Specialized skills^2	0.007	0.031**	0.022**
	(0.014)	(0.013)	(0.010)
Std. O*NET Balanced skill count	0.007	0.039	0.035
	(0.058)	(0.042)	(0.029)
Std. O*NET Balanced skill count^2	0.009	0.008	0.004
	(0.009)	(0.007)	(0.003)
Tenure in the business	0.175***	0.188***	0.134***
	(0.022)	(0.019)	(0.011)
Tenure in the business^2	-0.003***	-0.004***	-0.003***
	(0.001)	(0.001)	(0.000)
Number of person-year observations	2,465	2,465	5,938
Number of businesses	1,027	1,027	1,554
Adjusted R^2 (pseudo R^2 for Tobit)	0.065	0.177	0.167
Mean business revenue or earnings	215,578	1,088	1,088

Note: The models are those in table 9, with the same controls. Skill measures are standardized with the mean and standard deviation of the revenue sample so that a one-unit increase equals half a standard deviation. Mean revenue in column 1 excludes top-coded reports. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Appendix B: Other Approaches to the Endogeneity Problem

My estimates rely on selection on observables. Two other approaches could address the endogeneity problem discussed in section 5.3 without requiring me to observe the tastes that create it: individual fixed effects and an instrument for skills. In this appendix, I describe what each approach would address and why I rely on selection on observables in the main analysis instead.

The error term ϵ_{ijt} in equation 11 captures factors that shift the utility of option j but are not included in the observed regressors. Because BX_{it} and SX_{it} are built from workers' occupational histories, they depend not only on the skills each occupation uses but also on workers' choices of which occupations to enter and how long to stay in them. The main specification controls for experience, demographic characteristics, wealth, risk tolerance, cognitive ability, personality traits, and family background, but tastes I do not observe may still shape how workers allocate their years across occupations.

To show how unobserved tastes may affect the estimates, decompose ϵ_{ijt} as $\epsilon_{ijt} = u_{ij} + v_{ijt}$, where u_{ij} is time-invariant and v_{ijt} varies within a worker across years. The time-invariant component u_{ij} captures tastes that do not change over a worker's career, such as a desire for autonomy or an aspiration to own a business. A worker with such tastes may sort into occupations that use more business skills and stay in them longer in anticipation of entry. This creates a positive correlation between BX_{it} and u_{ij} for the business ownership options, so the estimated association between business skills and entry may reflect both the skills themselves and these tastes. The same issue applies to SX_{it}.

The time-varying component v_{ijt} captures changes in a worker's plans or circumstances over the career. For example, a worker who becomes interested in business ownership in mid-career may move into an occupation that requires more business skills before entering. Within-person changes in skills are then correlated with unobserved changes in the attractiveness of business ownership. Studies of returns to occupational tenure (Kambourov and Manovskii, 2009) and task-specific human capital (Gathmann and Schönberg, 2010) face the same problem. In these studies, workers stay longer in occupations that match them well, and because match quality changes when workers switch occupations, individual fixed effects cannot remove it.

B.1 Individual Fixed Effects for Entry into Business Ownership

Individual fixed effects would address the time-invariant component u_{ij}. The relationship between skills and entry into business ownership would be identified from within-person changes in BX_{it} and SX_{it} as workers move across occupations, so time-invariant tastes would no longer confound the estimates. Because the entry model is an MNL, this requires

a conditional logit model following Chamberlain (1980). This approach avoids imposing a distributional assumption on the individual effects and allows u_{ij} to be arbitrarily correlated with the observed covariates. A conditional rather than unconditional approach is necessary because estimating individual-specific intercepts in a nonlinear model introduces the incidental parameters problem.

However, individual fixed effects do not address the time-varying component v_{ijt}. A worker who becomes interested in business ownership in mid-career and then moves into occupations that use more business skills still generates within-person variation in BX_{it} that is correlated with her changing plans.

Moreover, individual fixed effects are less informative in my entry model. Because BX_{it} and SX_{it} accumulate steadily as a worker gains experience, little within-person variation in skills remains once experience is controlled for. Fixed effects are more useful for earnings, which are observed in both wage employment and business ownership. For earnings, the within-person variation comes mainly from moves between wage employment and business ownership rather than from changes in skills. The fixed effect absorbs a worker's overall earnings level, and the change in her earnings when she moves, compared across workers with different skills, shows how much more or less each skill pays in business ownership than in wage employment. I use this approach in appendix B.2.

B.2 Individual Fixed Effects for Earnings

Because weekly earnings are observed in both wage employment and business ownership, unlike business revenue, I can compare the returns to business skills and specialized skills in business ownership with their returns in wage employment for the same workers. My argument about the returns to different skills in business ownership and wage employment (section 3.2) also rests on this comparison. I argue that business skills are rewarded more in advanced business ownership than in wage employment and less in basic business ownership, and that wage employment is likely to reward specialized skills more than either type of business ownership (predictions 1 and 2). I use these estimated returns to check whether the entry choices reported in section 6.1 are consistent with how much each type of skill pays.

To make this comparison, I combine each worker's jobs in wage employment and in business ownership in one sample. I start with the earnings sample described in section 4.2, which contains one observation for each business and year with reported weekly earnings. I then add one observation for each wage job and year with reported weekly earnings for the 7,116 individuals in the entry sample. The resulting sample contains 191,590 person-year observations, of which 87,251 come from the 2,976 individuals who ever own a business.

I estimate the returns to skills in wage employment and in each type of business ownership

using the following model:

$$\ln w_{it} = \alpha_i + H'_{it}\pi_0 + \sum_{j\in\{B,A\}} D^j_{it}\left(\delta_j + H'_{it}\pi_j\right) + W'_{it}\beta + \epsilon_{it}, \tag{B1}$$

where $\ln w_{it}$ is the log of weekly earnings and α_i is the individual fixed effect. D^B_{it} and D^A_{it} equal one when the job is a basic or an advanced business, so wage employment is the omitted category. H_{it} contains BX_{it} and SX_{it}, their squares, their interactions with $Tenure_{it}$, $Tenure_{it}$ and its square, and actual and potential experience and their squares. $Tenure_{it}$ is tenure in the current job, whether it is a wage job or a business. All of these variables are measured at the time of the interview. π_0 captures how the skill and experience terms are related to earnings in wage employment, and π_B and π_A capture how these relationships differ in basic and advanced business ownership. Because the skill measures are interacted with tenure, the return to each skill can change as a worker stays longer in her job. W_{it} contains the time-varying controls of the revenue and earnings models: marital and cohabitation status, the young-child indicator, the number of past business ownership spells, and region and industry fixed effects. The individual fixed effect absorbs the remaining controls, which change little or not at all over a career, such as education. See table 1 for the full list of controls. Standard errors are clustered at the individual level.

Table B1 reports four versions of equation B1, which move step by step from a comparison across all workers to a comparison within each worker. Column 1 uses all workers and, instead of the individual fixed effect, includes the time-invariant controls. Column 2 keeps only workers who ever own a business, so that earnings in wage employment come from the same workers whose earnings in business ownership are observed. Column 3 adds the individual fixed effect, which absorbs time-invariant individual characteristics. Because a worker reports her earnings at every interview while she holds a job, businesses that last longer contribute more observations. To keep these businesses from dominating the estimates, column 4 repeats column 3 but weights each observation by the inverse of the number of observations in its job spell, so that each spell counts equally.

To interpret the estimates in column 4 of table B1, I calculate the average marginal effect of a half-standard-deviation increase in each skill measure reported in table B2. I do this separately for the two types of business ownership, using the observations of basic business owners in columns 1 to 3 and those of advanced business owners in columns 4 to 6. For each of these observations, I compute the return to a skill twice, once in business ownership and once as if the owner were a wage worker with the same skills and tenure, and then average each over the observations. The difference between the two averages shows how much more or less a skill pays the same workers in business ownership than in wage employment. The

table also reports these returns at zero, five, and ten years of tenure in the current job. I convert each estimate b to a percent change, $100(e^b - 1)$.

Business skills pay much less in basic business ownership than in wage employment (prediction 1), whereas in advanced business ownership the difference is small. In columns 1 and 2 of table B2, a half-standard-deviation increase in business skills predicts 7.0% higher earnings for basic business owners in wage employment but 16.2% lower earnings in basic business ownership. The difference in column 3 is statistically significant at every level of tenure, although it narrows as owners stay in their businesses, and the return in business ownership rises from −17.3% at the start of a business to −9.0% after ten years. In columns 4 and 5, the same increase is associated with 5.9% higher earnings in wage employment and 0.3% higher earnings in advanced business ownership. The difference in column 6 is not statistically significant at any level of tenure. The return in advanced business ownership also grows with tenure, from −2.7% at the start of a business to 5.0% after ten years, close to the 6.4% return in wage employment. These estimates do not show that business skills pay more in advanced business ownership than in wage employment. Still, compared with wage employment, they pay much more in advanced than in basic business ownership. As discussed in section 6.2, a possible reason is that owners with more business skills reinvest profits in their businesses rather than taking them home as earnings, which would lead earnings to understate the return to business skills in advanced business ownership.

The estimates do not support the prediction that specialized skills pay more in wage employment than in business ownership (prediction 2), because the differences in columns 3 and 6 have the opposite sign, although none of them is statistically significant. In columns 1 and 2 of table B2, a half-standard-deviation increase in specialized skills predicts 6.0% higher earnings for basic business owners in business ownership and 3.8% higher earnings for the same owners in wage employment. For advanced business owners, the same increase is associated with 7.0% higher earnings in business ownership and 2.8% higher earnings in wage employment.

Comparing columns 1 to 3 of table B1 also shows whether these differences in returns are driven by selection on time-invariant individual characteristics. This selection is part of the endogeneity problem discussed in section 5.3. The estimated coefficient on the interaction between business skills and basic business ownership changes little, from −0.298 in column 1 to −0.238 in column 3. Restricting the sample to workers who ever own a business and adding the individual fixed effect change it only slightly, so selection on time-invariant individual characteristics does not explain the low return to business skills in basic business ownership. In contrast, the estimated coefficient for advanced business ownership shrinks by more than a quarter, from −0.135 in column 1 to −0.097 in column 3, which might reflect some selection

on these characteristics in columns 1 and 2. The estimated coefficients on the interactions of specialized skills with both types of business ownership are positive in all three columns and become slightly smaller as the sample is restricted and the individual fixed effect is added.

Table B1: Coefficient Estimates for Log Weekly Earnings Using Individual Fixed Effects

	Dependent Variable: Log Weekly Earnings			
	(1)	(2)	(3)	(4)
Std. O*NET Business skills	0.153***	0.152***	0.058***	0.066***
	(0.011)	(0.018)	(0.017)	(0.021)
Std. O*NET Business skills^2	-0.010***	-0.010***	-0.003**	-0.004
	(0.001)	(0.002)	(0.002)	(0.003)
Std. O*NET Business skills × tenure in current job	0.002***	0.002***	0.003***	0.001
	(0.000)	(0.001)	(0.001)	(0.001)
Std. O*NET Business skills × 1 if basic business owner	-0.298***	-0.273***	-0.238***	-0.257***
	(0.055)	(0.055)	(0.053)	(0.059)
Std. O*NET Business skills^2 × 1 if basic business owner	0.019***	0.016**	0.011*	0.012*
	(0.007)	(0.007)	(0.006)	(0.007)
Std. O*NET Business skills × tenure in current job × 1 if basic business owner	0.005	0.004	0.005*	0.009***
	(0.003)	(0.003)	(0.003)	(0.003)
Std. O*NET Business skills × 1 if advanced business owner	-0.135**	-0.119**	-0.097*	-0.045
	(0.057)	(0.058)	(0.059)	(0.080)
Std. O*NET Business skills^2 × 1 if advanced business owner	-0.004	-0.006	-0.006	-0.012
	(0.007)	(0.007)	(0.007)	(0.009)
Std. O*NET Business skills × tenure in current job × 1 if advanced business owner	0.007***	0.007***	0.006**	0.007**
	(0.003)	(0.003)	(0.003)	(0.003)
Std. O*NET Specialized skills	0.043***	0.051***	0.028**	0.040***
	(0.006)	(0.011)	(0.011)	(0.013)
Std. O*NET Specialized skills^2	0.001	-0.001	-0.001	-0.002
	(0.001)	(0.002)	(0.001)	(0.002)
Std. O*NET Specialized skills × tenure in current job	-0.002***	-0.001**	-0.001**	-0.002**
	(0.000)	(0.001)	(0.001)	(0.001)
Std. O*NET Specialized skills × 1 if basic business owner	0.057*	0.050	0.039	0.017
	(0.033)	(0.033)	(0.032)	(0.037)
Std. O*NET Specialized skills^2 × 1 if basic business owner	0.002	0.004	0.005	0.005
	(0.007)	(0.007)	(0.006)	(0.006)
Std. O*NET Specialized skills × tenure in current job × 1 if basic business owner	0.002	0.002	0.001	0.002
	(0.002)	(0.003)	(0.002)	(0.003)
Std. O*NET Specialized skills × 1 if advanced business owner	0.046	0.036	0.034	0.015
	(0.043)	(0.043)	(0.042)	(0.053)
Std. O*NET Specialized skills^2 × 1 if advanced business owner	0.007	0.009*	0.007	0.013**
	(0.005)	(0.005)	(0.005)	(0.006)
Std. O*NET Specialized skills × tenure in current job × 1 if advanced business owner	-0.004*	-0.005*	-0.003	-0.003
	(0.002)	(0.002)	(0.002)	(0.003)
1 if basic business owner	-0.466***	-0.398***	-0.354***	-0.447***
	(0.133)	(0.132)	(0.129)	(0.149)
1 if advanced business owner	0.568***	0.609***	0.530***	0.675***
	(0.154)	(0.154)	(0.158)	(0.201)
Tenure in current job	0.065***	0.074***	0.075***	0.086***
	(0.001)	(0.002)	(0.002)	(0.003)
Tenure in current job^2	-0.002***	-0.002***	-0.002***	-0.002***
	(0.000)	(0.000)	(0.000)	(0.000)
Tenure in current job × 1 if basic business owner	0.096***	0.087***	0.072***	0.076***
	(0.011)	(0.011)	(0.011)	(0.013)
Tenure in current job^2 × 1 if basic business owner	-0.003***	-0.002***	-0.002***	-0.003***

Continued on next page

Table B1 (continued)

	Dependent Variable: Log Weekly Earnings			
	(1)	(2)	(3)	(4)
	(0.001)	(0.001)	(0.001)	(0.001)
Tenure in current job × 1 if advanced business owner	0.089***	0.080***	0.067***	0.078***
	(0.012)	(0.012)	(0.012)	(0.015)
Tenure in current job^2 × 1 if advanced business owner	-0.002***	-0.002***	-0.002***	-0.002***
	(0.000)	(0.000)	(0.000)	(0.001)
Actual experience	0.018***	0.024***	0.019***	0.017***
	(0.003)	(0.005)	(0.006)	(0.006)
Actual experience^2	-0.001***	-0.001***	-0.000***	-0.000***
	(0.000)	(0.000)	(0.000)	(0.000)
Actual experience × 1 if basic business owner	0.025	0.016	-0.002	0.001
	(0.016)	(0.017)	(0.015)	(0.018)
Actual experience^2 × 1 if basic business owner	-0.000	0.000	0.000	0.001
	(0.000)	(0.000)	(0.000)	(0.000)
Actual experience × 1 if advanced business owner	0.025	0.017	0.017	0.028
	(0.031)	(0.031)	(0.030)	(0.036)
Actual experience^2 × 1 if advanced business owner	-0.000	-0.000	-0.000	-0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Potential experience	-0.026***	-0.031***	-0.008**	-0.006
	(0.002)	(0.003)	(0.004)	(0.004)
Potential experience^2	0.000***	0.000***	0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Potential experience × 1 if basic business owner	-0.076***	-0.072***	-0.058***	-0.058***
	(0.012)	(0.012)	(0.011)	(0.012)
Potential experience^2 × 1 if basic business owner	0.000*	0.000*	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
Potential experience × 1 if advanced business owner	-0.140***	-0.134***	-0.132***	-0.172***
	(0.028)	(0.028)	(0.027)	(0.030)
Potential experience^2 × 1 if advanced business owner	0.002***	0.002***	0.002***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
Time-invariant baseline regressors	✓	✓		
Only workers who ever own a business		✓	✓	✓
Individual fixed effects			✓	✓
Weighted by inverse of spell-report frequency				✓
Mean dependent variable	6.157	6.055	6.055	6.055
Number of person-year observations	191,590	87,255	87,251	87,251
Adjusted R^2	0.322	0.292	0.382	0.364

Note: O*NET-based skill measures are standardized so that a one-unit increase equals half a standard deviation. See table 1 for the full list of covariates included. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

Table B2: Estimated Returns to Business Skills and Specialized Skills for Business Owners, in Business Ownership and in Wage Employment (Based on Column 4 of Table B1)

	Basic Business Ownership			Advanced Business Ownership		
	Wage Employment (1)	Business Ownership (2)	Difference (2)−(1) (3)	Wage Employment (4)	Business Ownership (5)	Difference (5)−(4) (6)
Std. O*NET Business skills						
Average	0.067***	-0.177***	-0.245***	0.058***	0.003	-0.055
	(0.021)	(0.062)	(0.059)	(0.015)	(0.059)	(0.059)
Tenure 0	0.066***	-0.190***	-0.257***	0.054***	-0.028	-0.082
	(0.021)	(0.062)	(0.059)	(0.016)	(0.064)	(0.063)
Tenure 5	0.070***	-0.143**	-0.213***	0.058***	0.010	-0.048
	(0.020)	(0.063)	(0.060)	(0.015)	(0.059)	(0.058)
Tenure 10	0.074***	-0.095	-0.169***	0.062***	0.049	-0.014
	(0.020)	(0.067)	(0.065)	(0.015)	(0.058)	(0.058)
Std. O*NET Specialized skills						
Average	0.038***	0.059	0.021	0.027***	0.068*	0.040
	(0.013)	(0.037)	(0.036)	(0.009)	(0.039)	(0.039)
Tenure 0	0.040***	0.057	0.018	0.034***	0.085**	0.051
	(0.013)	(0.037)	(0.037)	(0.011)	(0.042)	(0.043)
Tenure 5	0.032***	0.061*	0.030	0.026***	0.063*	0.038
	(0.012)	(0.037)	(0.037)	(0.009)	(0.038)	(0.039)
Tenure 10	0.023**	0.065	0.042	0.018**	0.041	0.024
	(0.012)	(0.040)	(0.041)	(0.009)	(0.039)	(0.040)
Business owner-years		10,380			5,938	

Note: Estimated returns correspond to a half-standard-deviation increase in each skill and are in log points. In columns 1 and 4, the same business owners are treated as wage workers with the same skills and tenure. Average rows use each owner's observed tenure in the current job. Standard errors clustered at the individual level are in parentheses. *p < .10; **p < .05; ***p < .01.

B.3 Instrumental Variables

An instrument may address both components of the error term. An effective instrument is required to shift the skills a worker accumulates but affect business ownership only through those skills. One candidate is the industrial composition of the county where a worker starts her career, in the spirit of Jara-Figueroa et al. (2018). Different occupations offered because of the circumstances of the local labor markets may affect the skills that workers accumulate. For example, a county whose economy is concentrated in manufacturing exposes workers to a different set of occupations, and with them a different path of skill accumulation than a county with a larger service sector or a concentration of professional industries. As a result, workers with similar initial characteristics may accumulate different skills because they begin their careers in different local labor markets.

I construct such an instrument for business skills, BX_{it}. For each county c, I use the 1975 County Business Patterns to calculate the share s_{kc} of employment in each industry k. I then measure how much each industry uses business skills, $\overline{BX}_k$, by assigning each occupation its business-skill score from section 4.4 and averaging across workers employed in industry k in the 1980 Census. The instrument for worker i, whose county of residence at labor market entry is $c(i)$, is:

$$Z_{ict}^{BX} = \sum_k s_{kc(i),1975} \cdot g_{kt} \cdot \overline{BX}_k, \tag{B2}$$

where g_{kt} is national cumulative employment growth in industry k from 1975 to year t. The instrument is higher for workers who started their careers in counties where industries that use more business skills accounted for a large share of employment and later grew nationally.

I construct a similar instrument for specialized skills:

$$Z_{ict}^{SX} = \sum_k s_{kc(i),1975} \cdot \overline{SX}_{kt}. \tag{B3}$$

Because specialized skills depend on each worker's target occupation, the industry score $\overline{SX}_{kt}$ cannot simply average occupation-level scores. Instead, it measures how much the occupations within industry k share their most important skills. For instance, in the construction industry, plumbers, electricians, and carpenters rely on many of the same skills, so working in any of these occupations builds skills that are useful in the others. In hospitals, nurses, accountants, and janitors share few of their most important skills. Workers who start their careers in counties dominated by industries of the first kind accumulate more transferable specialized skills. The score changes over time as the occupational mix within each industry changes, which I measure at each Census and American Community Survey benchmark from 1980 to 2015.

For either instrument to be valid, the industrial composition of a worker's starting county must affect business ownership only through the skills she accumulates. Two concerns remain. First, local economic conditions may directly affect business ownership decisions. For example, limited wage opportunities may push workers into business ownership regardless of their skills. Controlling for the county unemployment rate at labor market entry would absorb part of this channel. Second, workers may sort across counties based on their tastes for business ownership. Using the county of residence at labor market entry, rather than at the time of the business ownership decision, reduces this concern, as does restricting the sample to workers who stay in the same county during the period used to construct the instrument.

However, I face two problems in applying these two instruments. First, they are highly correlated, with a correlation of about 0.9 across counties. Industries whose occupations share business skills, such as finance and professional services, also rank high on how much their occupations share specialized skills. As a result, the instruments cannot separate the two types of skills, which the entry model requires. Second, the business-skill instrument is weak in the first stage with individual fixed effects, the setting of appendix B.2, where its only source of variation within a worker is national industry growth over time. Separating business from specialized skills would require a source of variation that shifts one type of skill without shifting the other, which I leave for future work.